Class-Imbalanced Graph Convolution Smoothing for Hyperspectral Image Classification

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***Abstract*— Graph convolutional network (GCN)-based methods for hyperspectral image (HSI) classification have received more attention due to its flexibility in information aggregation. However, most existing GCN-based methods in HSI community rely on capturing fixed K-hops neighbors for feature information aggregation, which ignores the inherent imbalance in class distributions and fails to achieve optimal feature smoothing through the graph convolution operator. It is unreasonable to apply a fixed K-hops strategy for feature smoothing in imbalanced classes, as class regions with rich contextual information and those with poor contextual information require to capture different hops neighbors to achieve the optimal feature smoothing. To address this issue, this article proposes a novel approach called class-imbalanced graph convolution smoothing (CIGCS) for HSI classification, which achieves adaptive feature smoothing for imbalanced class regions. First, we construct a semantic block-diagonal graph structure that describes imbalanced semantic class regions by considering label connectivity and spectral Laplacian regularizer. Second, we develop the CIGCS technique to adaptively aggregate neighbor information for imbalanced class regions based on the decreasing Euclidean distance of samples within each bock-diagonal structure from the perspective of oversmoothing. The choice of adaptive neighbors can be guaranteed by a theoretical upper bound. Finally, the obtained optimal smoothed features are fed into the logistic regression to achieve good classification results. The proposed CIGCS method is evaluated on three real HSI datasets to demonstrate its superiority compared to some popular GCN-based methods.**

***Index Terms*— Block-diagonal graph structure, class-imbalanced feature smoothing, graph convolutional network (GCN), hyperspectral image (HSI) classification.**

NOMENCLATURE

Symbol	Definition
$\mathbf{U}$	Orthonormal eigenvectors.
$\mathbf{L}$	Symmetric positive semidefinite Laplacian matrix.
$\mathbf{D}$	Degree formation of nodes.
$\boldsymbol{g}_\theta$	Function of the eigenvalues.
$\Lambda_{\max}$	Large eigenvalues of matrix $\mathbf{L}$.
I	Identity matrix.
$\mathbf{A}$	Adjacency similarity graph.
$\mathbf{X}$	Initial feature information.
$\Theta^{(k)}$	Weight parameter from the kth layer.
$\mathbf{Z}$	Prediction result.
$\boldsymbol{S}$	Spatial–spectral unified adaptive probability graph (SSAPG).
$\boldsymbol{P}$	Guidance matrix based on expert prior knowledge.
λ, β	Balanced parameters.
Q	Label guidance matrix.
q_U	Label distribution of unlabeled samples.
$\text{rank}(L_s)$	Low-rank restriction.
$\| \boldsymbol{S} \|_{\langle C \rangle}$	C-block-diagonal regularizer.
$\text{intra}(b_r)$	Distance of samples within rth block-diagonal structure.
K_r	Optimal hops of neighbors for optimal feature smoothing.
ε	Parameter of indicating smoothing effect.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) are obtained by sampling the reflected solar radiation from the Earth's surface across a range of wavelengths between 0.3 and 2.5 μm, encompassing the visible, near-infrared, and midinfrared regions with narrow contiguous bands [1], [2]. With their rich and detailed spectral information, HSIs offer the ability to discriminate objects of interest, particularly those with similar spectral characteristics. Compared with RGB and multispectral data, HSIs provide a more detailed and accurate classification of the materials on the Earth's surface [3]. However, the complex spatial–spectral characteristics, such as high spectral variation in intraclass pixels and high spectral similarity in interclass pixel, bring challenges for HSI classification [4]. For handling these challenges, researchers have proposed feature smoothing techniques to alleviate spectral variation by aggregating neighboring feature information. Representative methods include convolutional neural networks (CNNs) and graph convolutional network (GCN)-based methods.

Over the past decades, various classification methods based on handcrafted features have been proposed for HSI classification. For instance, spatial–spectral principal component analysis (PCA) [5] and spatial–spectral independent component analysis (ICA) [6] are traditional classification methods

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that aim to transform the original high-dimensional feature space into a lower dimensional feature space to distinguish feature information between semantic classes. In addition, methods based on subspace representation learning have been proposed to find the subspace representation of each sample by using self-expression ability from its own subspace to represent itself. Representative methods include sparse presentation [7], [8] or low-rank representation [9], [10]. Although these methods have been proven effective in HSI classification, they still lack powerful data fitting and representation ability.

Inspired by the success of deep learning-based methods, various techniques in the field of HSI classification [11]. The reason behind this is that deep learning techniques integrate feature extraction with downstream tasks, creating a unified framework that automatically learns more suitable features to handle different situations effectively [12]. CNN is a classical representative method in deep learning that achieves regular feature smoothing, and CNN-based methods can be broadly categorized into 1-D CNN, 2-D CNN, and 3-D CNN. For example, Chen et al. [13] proposed the deep belief network (DBN) [13] method for HSI classification, which utilizes a 1-D CNN network to extract effective spectral features. The 1-D CNN focuses more on spectral information, while 2-D CNN emphasizes spatial information. Consequently, a hybrid architecture combining 1-D and 2-D CNN layers is proposed to learn both spectral and spatial information for achieving effective HSI classification [14], [15]. In another example, Wang et al. [16] proposed the adaptive spectral–spatial multiscale contextual network (ASSMN) for HSI classification, leveraging the convolutional long short-time memory model. Recognizing that different spectral bands and spatial positions exhibit varying discriminative abilities in HSIs, Hang et al. [17] proposed attention-aided CNNs for spectral–spatial HSI classification. To further explore the simultaneous effectiveness of the spectral and spatial information in HSI classification, 3-D CNN models [18], [19] were developed to directly extract spatial–spectral features from raw HSI data. For example, Qi et al. [20] proposed a method entitled global–local 3-D convolutional transformer network (GTCT), where 3-D convolution is integrated into a dual-branch transformer to capture global–local associations in both spectral and spatial domains of HSIs. HybridFormer [21] explored the richer semantic features by designing the token generator by embedding the multigranularity semantic tokens and using the spatial–spectral attention mechanism. In fact, semantic features for the hybrid transformer can be obtained by the fixed depthwise convolution size, which lacks adaptability for the feature extraction of semantic class. In addition, Paoletti et al. [22] devised a method that automatically designs and optimizes CNNs for HSI classification using channel-based attention mechanisms, incorporating 1-D and spectral–spatial 3-D classifiers to handle the substantial information contained in HSIs from different perspectives.

However, methods based on deep learning typically utilize regular convolutional kernels to extract multiscale features with a large number of hyperparameters. This approach struggles to achieve good generalized performance for irregular distributions of imbalanced class regions in the HSI community, due to the complex spectral characteristics. To address this issue, the semi-supervised GCN [23], [24] method has been applied to HSI classification. The GCN leverages graph connectivity and the feature smoothing through irregular graph convolution operations. Numerous GCN-based methods have been proposed for HSI classification. For example, the spectral–spatial GCN (S^2GCN) [25] method and the spatial–spectral unified adaptive probability GCN (SSAPGCN) [26] combine intrinsic spectral information and local spatial window information for HSI classification, exploiting the strengths of GCN models. However, these methods rely on a local graph structure, which overlooks comprehensive contextual information from a global perspective and limit inference performance. To capture rich contextual information with global high-order consistency, researchers have developed global GCN-based models such as global consistent GCN (GCGCN) [27] and nonlocal GCN [28]. Though these models achieve good classification performance, they fail to consider the spectral–spatial correlations at multiple scales. To address this limitation, the multiscale dynamic GCN (MDGCN) [29], the context-aware dynamic GCN (CAD-GCN) [30], and the multiscale receptive fields graph attention neural network (MRGAT) [31] were proposed, utilizing superpixel-based graph structures to achieve superpixel-level neighbor information aggregation instead of pixel-level neighbor information aggregation. However, pixel- and superpixel-level data representations are incompatible in a single network. To overcome this challenge, the CNN-enhanced GCN (CEGCN) [32] method was proposed, enabling collaboration between CNN (with pixel-level representation) and GCN (with superpixel-level representation) in a single network. Capitalizing on the strengths of both CNN and GCN, Zhou et al. [33] proposed the attention multihop graph and multiscale convolutional fusion network (AMGCFN) for HSI classification, which utilizes multiple feature fusion models. In addition, the multifeature fusion network (MFGCN) [34] and the weighted feature fusion of CNN and graph attention network (WFCG) [35] methods were proposed to explore the feature fusion between GCN and CNN methods in pixel and superpixel levels. They try to capture long-range information with the help of the superpixels, but it is unreasonable to adopt the fixed graph convolution to achieve feature extraction. Next, the multiadaptive receptive field-based graph neural framework (MARP) [36] method was proposed to suppress the impact of noise nodes and automatically explore an adaptive receptive field for HSI classification. While the above GCN-based methods are effective for transductive learning, they cannot handle out-of-sample data. To address this issue, the minibatch GCN (MiniGCN) and its improved end-to-end fusion network (FuNet) [37], incorporating both MiniGCN and CNN, were proposed to achieve HSI classification.

Although the aforementioned GCN-based methods have achieved good classification performance, they only consider the fixed-length propagation paradigm for feature smoothing

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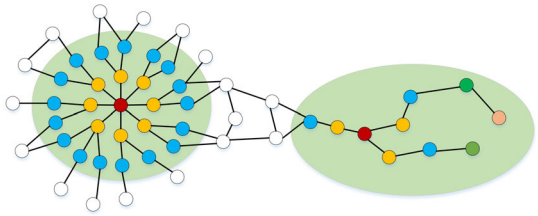


Fig. 1. Graph nodes (red node) in dense and sparse class regions have different feature smoothing effects with different graph structures. The red nodes denote the core node, orange nodes denote the one-hop neighbors, blue nodes denote two-hops neighbors, green nodes denote the three-hop neighbors, and pinkish-orange nodes denote the four-hop neighbors. The left red resides in the center of the rich contextual class region, while the right red one has the limited neighbors in the poor contextual class region. In other words, the number of capturing $K-$hops neighbors to reach an optimal level of smoothness is rather different for the red nodes. Ideally, poorly connected nodes (e.g., the red node on the right) need large iteration numbers to efficiently gather high-order information from other nodes, while densely connected nodes (e.g., the red node on the left) would require fewer hops to gather low-order information to achieve optimal feature smoothing.

via graph convolution in HSI classification. Note that the MARP method is also adopting fixed graph convolution layers (four layers) to aggregate multiscale feature information for each HSI. This propagation paradigm aggregates uniform $K-$hops neighbors for all semantic classes to achieve feature smoothing. It may not be reasonable to adopt fixed $K-$hops neighbors to achieve feature smoothing because of the imbalanced class distribution in HSIs. Intuitively, larger class regions with rich contextual information tend to have a dense connected graph structure, while smaller class regions with poor contextual information usually have a sparse graph structure due to the limited semantic class information. Moreover, the choice of $K-$hops neighbors in pursing optimal feature smoothing of GCN-based methods is closely related to the graph structure. Thus, class-imbalanced regions should capture adaptive neighbors to achieve optimal feature smoothing for each semantic class because of its different graph structure with different contextual information. The graphical explanation is shown in Fig. 1.

In order to address the aforementioned issues, this article proposes a novel method called class-imbalanced graph convolution smoothing (CIGCS) to handle the influence of fixed-length information propagation on imbalanced class distributions in HSI classification. Specifically, we first construct the semantic block-diagonal structure to describe the imbalanced class regions and obtain the appropriate graph connectivity structure within each imbalanced class region. This is based on the underlying assumption that there is semantic connectivity between any pair of pixels within each class, which helps reveal the distributions of imbalanced class regions. Next, we proposed to perform adaptive feature smoothing by aggregating adaptive hops neighbors using graph convolution operations, instead of using the fixed-length information propagation paradigm commonly used in GCN-based methods for imbalanced class regions. This is done based on the different block-diagonal structures reflected by each imbalanced region, where each block-diagonal structure represents a semantic class. The contributions of this article can be summarized as follows.

1) The CIGCS method is proposed for HSI classification, which takes the imbalanced class distributions into account in feature smoothing by capturing adaptive neighboring information at the class level and achieves the effective feature smoothing rather than relying on fixed-length information propagation paradigm used by most GCN-based methods.

2) The block-diagonal structure with label connectivity and low-rank restriction is proposed to describe the imbalanced class distribution, and the class-imbalanced graph convolution with the decreasing Euclidean distance within intraclass and oversmoothing measurement is proposed to achieve the adaptive feature smoothing of imbalanced class regions.

3) Compared to the other popular GCN-based methods, the determined adaptive hops neighbors obtained by the proposed CIGCS method can be guaranteed by a theoretical upper bound in pursuing feature smoothing. Moreover, comprehensive experiments demonstrate the superiority of the CIGCS method on HSI classification.

II. Related Work

The information propagation of the proposed CIGCS method is motivated by the vallina GCN [24] method and the decoupled idea between feature aggregation and feature transformation is motivated by the SGC [38] method. The basic idea of GCN and SGC methods is close to the proposed CIGCS method. Therefore, the basic theory of the above two methods can be introduced as follows.

A. GCN

GCN is a semi-supervised classification method, which learns hidden layer representations that encode both local graph structure and features of nodes. Spectral convolution on graphs is the origin of the GCN model, which relies on the spectral decomposition of graph Laplacians using a graph Fourier analysis [39]. Let $\boldsymbol{x} \in \mathbb{R}^N$ be a signal defined on the vertices of the graph, and the graph convolution operation between the signal $\boldsymbol{x}$ and the filter $\boldsymbol{g}$ is defined as

$$\boldsymbol{g} * x = \mathbf{U}\left(\left(\mathbf{U}^{\mathrm{T}} g\right) \odot \left(\mathbf{U}^{\mathrm{T}} x\right)\right) = \mathbf{U} g_\theta \mathbf{U}^{\mathrm{T}} x \tag{1}$$

where $\mathbf{U} \in \mathbb{R}^{N \times N}$ represents the orthonormal eigenvectors of normalized Laplacian matrix $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, $\boldsymbol{U}^{\mathrm{T}} x$ represents the graph Fourier transform that is equivalent to transform the signal $\boldsymbol{x}$ to the graph domain, $\odot$ indicates the Hadamard product (which is an elementwise product), and g_θ can be understood as a function of the eigenvalues of $\mathbf{L}$, i.e., $g_\theta(\Lambda)$. The multiplication with the eigenvector matrix $\mathbf{U}$ in (1) is computationally expensive with $O(N^2)$ and the eigendecomposition of the Laplacian matrix $\mathbf{L}$ may be prohibitively expensive for large graphs. Thus, to circumvent this problem, Hammond et al. [40] proposed to approximate $g_\theta(\Lambda)$ by a truncated expansion in terms of Chebyshev polynomials $\mathrm{T}_k(x)$ up to K order

$$\boldsymbol{g}_{\theta'}(\Lambda) \approx \sum_{k=0}^{\mathrm{K}} \theta' \mathrm{T}_{\mathrm{k}}\left(\tilde{\Lambda}\right) \tag{2}$$

where $\tilde{\Lambda} = (2\Lambda/\Lambda_{\max}) - I$ is a rescaled version, $\Lambda_{\max}$ denotes the largest eigenvalue of $\mathbf{L}$, and the Chebyshev polynomials

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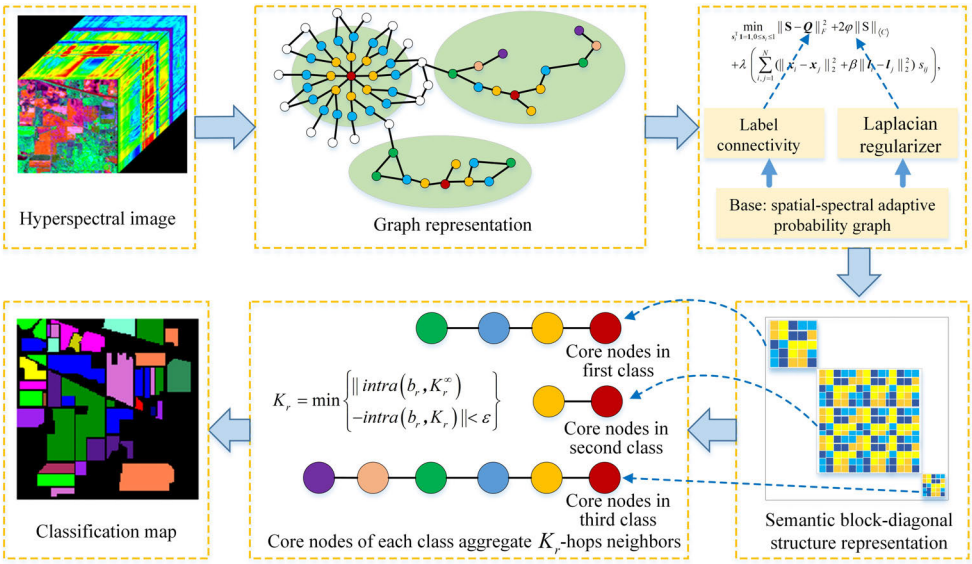


Fig. 2. Flowchart of the proposed CIGCS model. The circle in the graph representation stage denotes graph nodes. Different nodes denote different hops neighbors. The red circles denote the core nodes, orange circles denote the one-hop neighbors of the core nodes, blue circles denote the two-hop neighbors, green circles denote the three-hop neighbors, pinkish-orange circles denote the four-hop neighbors, violet circles denote the five-hop neighbors, and so on. The nodes within each block-diagonal structure b_r (each semantic class) are able to aggregate k_r-hop neighbors to achieve optimal feature smoothing.

are recursively defined as $\mathrm{T}_k(x) = 2x\mathrm{T}_{k-1}(x) - \mathrm{T}_{k-2}(x)$ with $\mathrm{T}_0(x) = 1$ and $\mathrm{T}_1(x) = x$. The definition of the graph convolution of the signal $\boldsymbol{x}$ with the filter g_θ can be rewritten as

$$\boldsymbol{g}_{\theta'} * x \approx \sum_{k=0}^{K} \theta' \mathrm{T}_k\left(\tilde{L}\right) * x \tag{3}$$

where $\tilde{L} = (2L/\Lambda_{\max}) - \boldsymbol{I}$. From (3), the expression is now K-localized version in the Laplacian that it depends only on the maximum K steps away from the central node. Defferrard et al. [41] proposed the K-localized convolution to define graph convolution. Next, Kipf and Welling [24] further proposed to approximate $\Lambda_{\max} \approx 2$ during training, and (3) can be simplified as

$$\boldsymbol{g}_{\theta'} * x \approx \theta'_0 x + \theta'_1(\boldsymbol{L} - \boldsymbol{I})x = \theta'_0 x - \theta'_1 \boldsymbol{D}^{-1/2}\boldsymbol{A}\boldsymbol{D}^{-1/2}x \tag{4}$$

where the two parameters θ'_0 and θ'_1 can be shared over the whole graph. In practice, the number of parameters can be simplified to further address feature overfitting during training, and (4) can be rewritten as

$$\boldsymbol{g}_\theta * x \approx \theta\left(\boldsymbol{I} + \boldsymbol{D}^{-1/2}\boldsymbol{A}\boldsymbol{D}^{-1/2}\right)x \tag{5}$$

where $\theta = \theta'_0 = -\theta'_1$. The graph convolution can be generalized to a signal with matrix version $X \in \mathbb{R}^{N\times C}$ with C-dimensional feature vector and F filters or feature maps as

$$\boldsymbol{Z} = \boldsymbol{I} + \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1/2}\boldsymbol{X}\Theta \tag{6}$$

where $\Theta \in \mathbb{R}^{C\times F}$, $\boldsymbol{I} + \boldsymbol{D}^{-1/2}\boldsymbol{A}\boldsymbol{D}^{-1/2} \to \tilde{\boldsymbol{D}}^{-1/2}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-1}/2$ is a renormalization trick to avoid numerical instabilities and vanishing gradients, $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}$, and $\tilde{\boldsymbol{D}}_{ii} = \sum_j \tilde{\boldsymbol{A}}_{ij}$. Finally, the two-layer GCN for semi-supervised classification model can be defined as

$$\boldsymbol{Z} = f(\boldsymbol{X}, \boldsymbol{A}) = \sigma\left(\hat{A}\mathrm{ReLU}\left(\hat{A}X\Theta^{(0)}\right)\Theta^{(1)}\right) \tag{7}$$

where σ is an activation function.

B. SGC

The GCN method derives inspiration primarily from recent deep learning approaches and may inherit unnecessary complexity. Consequently, SGC is proposed to reduce the redundant computation by removing nonlinearities and collapsing weight matrices between consecutive layers. In fact, the SGC method decouples feature propagation and feature transformation in the GCN method, whose decoupled way is beneficial to reduce the redundant computation. Hypothesize that the nonlinearity between GCN layers is not important, and the major benefit derives from the local averaging of node features. Therefore, the SGC model removes the nonlinear transition functions between each layer and only keeps the final softmax function to obtain the probability value of outputs. In fact, the SGC model is linear and has the same increased receptive field of the K-layer GCN

$$\boldsymbol{Z} = \sigma\left(\hat{\boldsymbol{A}}\ldots\hat{\boldsymbol{A}}\hat{\boldsymbol{A}}\Theta^{(1)}\Theta^{(2)}\ldots\Theta^{(K)}\right). \tag{8}$$

For simplifying the notation, (8) can be rewritten as

$$\boldsymbol{Z}_{\mathrm{SGC}} = \sigma\left(\hat{\boldsymbol{A}}^K\boldsymbol{X}\Theta\right). \tag{9}$$

In fact, the SGC model consists of a fixed smoothing component $\hat{\boldsymbol{A}}^K\boldsymbol{X}$, which is essentially equivalent to a feature preprocessing step. Then, the classification results can be obtained by straightforward multiclass logistic regression on the preprocessed smoothed features $\hat{\boldsymbol{A}}^K\boldsymbol{X}$.

III. Proposed Method

In this section, Fig. 2 illustrates the flowchart of the proposed CIGCS model, which consists of two critical modules: the semantic block-diagonal structure and CIGCS. First, the semantic block-diagonal graph is constructed based on the spatial–spectral unified adaptive probability graph

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(SSAPG) model. This construction incorporates further restrictions, including label connectivity and Laplacian regularizer, to preserve intraclass connections and suppress interclass connections from both local and global perspectives. Next, the adaptive graph convolution operator is applied to imbalanced classes using the semantic block-diagonal graph structure, which describes the distribution of imbalanced class regions. This operator aggregates adaptive neighboring information to achieve optimal feature smoothing. The selection of optimal neighbors for each class is guaranteed by considering the difference in the Euclidean distance between samples within the compact block-diagonal structure and its theoretical upper bound, rather than approximating infinite-hop neighbors. In summary, the combination of semantic block-diagonal structure and CIGCS ensures the good classification performance. Besides, for the convenience of reading, these symbols in this article are listed in the Nomenclature.

A. Block-Diagonal Structure

The block-diagonal structure aims at constructing the connectivity graph within each class to describe the distribution of imbalanced class regions, which can preserve the compact intraclass connections and suppress the interclass connections. We expect the semantic block-diagonal graph structure can describe the connectivity relationships within each class region in HSI instead of only considering sparsity of the graph structure such as common GCN-based methods. The block-diagonal structure is based on the SSAPG structure, which aims at exploring the local connectivity of each pair of homogeneous pixels [26]. The mathematical formula of the SSAPG model can be written as

$$\min_{s_i^{\mathrm{T}}1=1,0\le s_i\le 1} \| \mathrm{S}-\mathrm{P} \|_F^2 + \lambda\left(\sum_{i,j=1}^{N}\left(\| \boldsymbol{x}_i-\boldsymbol{x}_j \|_2^2+\beta \| \boldsymbol{l}_i-\boldsymbol{l}_j \|_2^2\right)s_{ij}\right) \tag{10}$$

where $S \in \mathbb{R}^{N\times N}$ denotes the SSAPG structure used to capture the pairwise probability connections within the same category, $P \in \mathbb{R}^{N\times N}$ is the guidance matrix based on expert prior knowledge that constrains the SSAPG, $x_i \in \mathbb{R}^{1\times d}$ denotes the ith pixel, $l_i \in \mathbb{R}^{1\times 2}$ denotes the spatial position coordinate of the ith pixel, and λ and β denote the balanced parameters. Although the SSAPG method preserves the local structure alleviating the spectral information variation of intraclass samples to a certain degree, it cannot describe the distribution of imbalanced class regions via local semantic connectivity relationships within each class. In fact, good semantic connectivity relationships can describe the distribution of semantic classes to a certain degree. The label connectivity utilizes the label distribution to improve the dependence of samples within imbalanced classes. To address this, we propose to utilize the label connectivity of a homogenous graph to improve intraclass compactness. The ideal label distribution within each class only considers the intraclass pixels, alleviating the spectral variation of intraclass samples from a label connectivity perspective. Label connectivity improves the connectivity relationships of intraclass samples, increasing the dependence on each class and reducing the influence of spectral variation. The SSAPG model with the guidance of label connectivity can be written as

$$\min_{s_i^{\mathrm{T}}1=1,0\le s_i\le 1} \| \mathrm{S}-\boldsymbol{Q} \|_F^2 + \lambda\left(\sum_{i,j=1}^{N}\left(\| \boldsymbol{x}_i-\boldsymbol{x}_j \|_2^2+\beta \| \boldsymbol{l}_i-\boldsymbol{l}_j \|_2^2\right)s_{ij}\right) \tag{11}$$

where $\boldsymbol{Q}=\boldsymbol{q}\cdot\boldsymbol{q}^{\mathrm{T}} \in \mathbb{R}^{N\times N}$ denotes the label guidance matrix obtained by label distribution and $\boldsymbol{q}=[\boldsymbol{q}_L;\boldsymbol{q}_U] \in \mathbb{R}^{N\times C}$ refers to the probability of labeled and unlabeled samples belonging to their corresponding true class, in which $L+U=N$, with L and U the number of the training samples and testing samples, respectively. $\boldsymbol{q}_L \in \mathbb{R}^{L\times C}$ denotes the given label distribution from prior knowledge, and $\boldsymbol{q}_U \in \mathbb{R}^{U\times C}$ denotes the label distribution of unlabeled samples obtained through the classical label propagation algorithm [42]. $\boldsymbol{q}_U$ is defined as

$$\boldsymbol{q}_U = \boldsymbol{S}_{UL}\cdot\boldsymbol{q}_L \tag{12}$$

where $\boldsymbol{S}_{UL} \in \mathbb{R}^{U\times L} \subset \boldsymbol{S}$ denotes the spatial–spectral probability graph between labeled samples and unlabeled samples belonging to the same class.

Equation (12) aims at assign label information to unlabeled samples based on their labeled neighbors. The label guidance matrix $\boldsymbol{Q} \in \mathbb{R}^{N\times N}$ preserves the intraclass connectivity, compensating for the lack of sparseness in the SSAPG structure $\boldsymbol{S}$. The original guidance matrix $\boldsymbol{P} \in \mathbb{R}^{N\times N}$ only preserves the sparseness of intraclass connectivity by using a spatial–spectral distance restriction, which is influenced by spectral variation due to only considering spatial–spectral restrictions. It cannot guarantee effective intraclass connectivity or describe the connectivity relationships of intraclass samples within each class region. Although the label connectivity guidance matrix preserves the intraclass connectivity to some degree, interclass connections in the graph structure are inevitably introduced under the restriction of label connectivity due to the interclass similarity of spectral information of imbalanced class regions. Next, we further consider interclass sparsity to improve the model by suppressing interclass connectivity relationships in the graph structure. Since noisy connections among interclass samples in the graph structure usually exhibit high-rank characteristics [43], [44], we consider the low-rank characteristic to remove the noisy interclass connections. Given a graph with an affinity matrix $\boldsymbol{S}$, if the rank of Laplacian matrix $\boldsymbol{L}_s$ with $\boldsymbol{L}_s=\boldsymbol{D}-\boldsymbol{S}$ is $\mathrm{rank}(\boldsymbol{L}_s)=N-C$, the affinity matrix $\boldsymbol{S}$ is able to remove noisy interclass connections while preserving the low-rank property in the graph structure to a certain degree. Since the restriction $\mathrm{rank}(\boldsymbol{L}_s)=N-C$ is a complex nonlinear low-rank restriction compared to full-rank property, it is difficult to directly achieve. However, it can be satisfied using the block-diagonal regularizer according to [45]. Specifically, the multiplicity of C of the eigenvalue zero of the Laplacian matrix $\boldsymbol{L}_s$ is equal to the number of connected components in the affinity matrix $\boldsymbol{S}$, and then, the C-block-diagonal regularizer is defined as $\| \boldsymbol{S} \|_{\langle C\rangle}=\sum_{i=0}^{C}\Lambda_i(\boldsymbol{L}_s)$. The Laplacian matrix $\boldsymbol{L}_s$ has the following important C-block-diagonal property, and Λ_i denotes the eigenvalues. Ideally,

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$\| \boldsymbol{S} \|_{\langle C \rangle} = 0$ is equivalent to the affinity matrix $\boldsymbol{S}$ possessing the C-block-diagonal property. Therefore, $\| \boldsymbol{S} \|_{\langle C \rangle}$ can be regard as the block-diagonal regularizer that forces the affinity matrix $\boldsymbol{S}$ to suppress noisy interclass connection. Consequently, to remove noisy interclass connections in the graph structure, the improved block-diagonal representation can be reformulated as

$$\min_{s_i^{\mathrm{T}} 1=1, 0 \le s_i \le 1} \| \mathrm{S} - \boldsymbol{Q} \|_F^2 + 2\varphi \| \mathbf{S} \|_{\langle C \rangle} + \lambda \left(\sum_{i,j=1}^{N} \left(\| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 + \beta \| \boldsymbol{l}_i - \boldsymbol{l}_j \|_2^2 \right) s_{ij} \right) \tag{13}$$

where the parameter φ is a balanced parameter. In fact, the noisy interclass connections in the similarity matrix $\boldsymbol{S}$ can be well removed when the restriction $\| \boldsymbol{S} \|_{\langle C \rangle} \to 0$, i.e., the sum of the C smallest eigenvalues is equal to zero. In other words, the noisy connection in the similarity matrix $\boldsymbol{S}$ can be alleviated by maintaining a connectivity graph with a low-rank property. Based on (13), the final block-diagonal structure with local and global characteristics preserved intraclass connectivity and suppressed interclass connectivity, which effectively describes the intrinsic distribution of imbalanced regions.

B. Class-Imbalanced Graph Convolution

In fact, the class-imbalanced region reflected in graph structure leads to an imbalanced topology. Given the aforementioned semantic block-diagonal structure describing the distribution of imbalanced class regions, our goal is to develop adaptive graph convolution techniques that can aggregate information from adaptive neighboring nodes to achieve optimal feature smoothing for these imbalanced regions. However, most existing GCN-based methods typically capture fixed K-hops neighbors (often $K = 2$) to smooth the spectral features for HSI classification, making it difficult to achieve optimal feature smoothing for imbalanced class regions. Generally, large-class regions with dense connectivity only need to capture a few hops neighbors to achieve satisfying feature smoothing and capturing high-order neighbors will cause oversmoothing. On the other hand, small-class regions with sparse connectivity compared to the dense connectivity of large-class regions usually require capturing high-order neighbors to achieve better feature smoothing, and capturing only a few hops neighbors will result in undersmoothing. As a result, it is challenging to achieve effective feature smoothing within each class using graph convolution operations. Intuitively, we expect to perform adaptive graph convolution operation for class-imbalanced regions to achieve optimal feature smoothing within each semantic class (block structure). In other words, the class-imbalanced regions should capture adaptive hops neighbors to achieve optimal feature smoothing. The relevant works utilize the multiscale information to achieve effective feature aggregation in graph convolution. For example, the MDGCN and MARP methods use the multiscale information to compensate for the shortcoming of the aggregation strategy of fixed K-hops neighbors in graph convolution. However, they only consider the feature smoothing via graph convolution at the node level and ignore the effective feature smoothing of imbalanced semantic classes at the class level. In fact, the feature smoothing of all semantic classes is beneficial to alleviate spectral variation of HSIs. Thus, we proposed the class-imbalanced graph convolution to perform the adaptive feature smoothing at class level, especially capturing high-order neighbors for feature smoothing.

To accomplish this goal, we propose to utilize the intraclass distance [46] to measure the effect of feature smoothing. The theoretical basis for this approach comes from [22], which demonstrates the intrinsic property of graph convolution is a special form of Laplacian smoothing. The intraclass distance, which represents the compactness of cluster partition, is first introduced as follows:

$$\mathrm{intra}(C) = \frac{1}{|C|} \sum_{i=1}^{|C|} \frac{1}{|c_i|(|c_i|)} \sum_{\boldsymbol{x}_i, \boldsymbol{x}_j \in c_i} \| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 \tag{14}$$

where c_i represents the ith class, $|c_i|$ represents the number of samples in the ith class, and $|C|$ represents the number of all classes. Inspired by intraclass distance, the distance of samples within each block-diagonal structure can be written as $\mathrm{intra}(b_r)$ where b_r denotes the rth block-diagonal structure. Generally, the distance (i.e., Euclidean distance) of spectral feature information will decrease with deeper Laplacian smoothing until the occurrence of oversmoothing. Therefore, to achieve optimal feature smoothing for each imbalanced region described by the block-diagonal structure b_r in the graph, we propose using oversmoothing to find the optimal K_r- hops neighbors for b_r, and it can be defined mathematically as

$$K_r(b_r) = \min \left\{ \| \mathrm{intra}\left(b_r, K_r^{\infty}\right) - \mathrm{intra}(b_r, K_r) \| < \varepsilon \right\} \tag{15}$$

where K_r denotes the optimal K_r-hops neighbors for each block-diagonal structure b_r, achieving optimal feature smoothing, ε denotes a parameter with a small value indicating strong smoothing effects, and $\mathrm{intra}(b_r, K_r)$ denotes the distance of samples smoothed by K_r-hops neighbors. Similarly, $\mathrm{intra}(b_r, K_r^{\infty})$ denotes the distance of samples smoothed by approximately infinite hops neighbors. The mathematical formula $\mathrm{intra}(b_r, K_r)$ can be written as

$$\mathrm{intra}(b_r, K_r) = \frac{1}{|b_r|(|b_r|)} \sum_{\boldsymbol{x}_i, \boldsymbol{x}_j \in b_r} \| \hat{x}_i^{K_r} - \hat{x}_j^{K_r} \|_2^2 \tag{16}$$

where $\hat{\boldsymbol{x}}_i^{K_r} \in \hat{\boldsymbol{X}}_{b_r}$ denotes the smoothed features aggregated from K_r-hops neighbors for the block-diagonal structure region b_r. $\hat{\boldsymbol{X}}_{b_r}$ can be written as

$$\hat{\boldsymbol{X}}_{b_r} = \hat{\boldsymbol{S}}_{b_r}^{K_r} \cdot \boldsymbol{X}_{b_r} \tag{17}$$

where $\hat{\boldsymbol{X}}_{b_r}$ denotes the optimal smoothed feature within block structure b_r and $\hat{S}_{b_r}^{K_r} = \tilde{\boldsymbol{D}}^{-1} \tilde{\boldsymbol{S}}_{b_r}^{K_r}$ denotes the graph structure for block-diagonal structure b_r, capturing the optimal K_r-hops neighbors in which $\tilde{\boldsymbol{S}}_{b_r}^{K_r} = \boldsymbol{S}_{b_r}^{K_r} + \boldsymbol{I}$ and $\boldsymbol{X}_{b_r}$ denotes the original feature within the block b_r. Any samples belonging to the block-diagonal structure region b_r will aggregate the feature information from K_r- hops neighbors to achieve optimal feature smoothing. Similarly, $\mathrm{intra}(b_r, K_r^{\infty})$ can be obtained via the multiplication of a block graph structure $\hat{\boldsymbol{S}}_{b_r}^{\infty}$ with original feature $\boldsymbol{X}_{b_r}$, that is, $\hat{\boldsymbol{S}}_{b_r}^{\infty} \cdot \boldsymbol{X}_{b_r}$.

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Based on (15), the basic idea is to consider adaptive smoothed features within each block-diagonal structure b_r by aggregating optimal K_r-hops neighbors information instead of leading to oversmoothing phenomenon with infinite hops neighbors through the restriction of parameter ε. When the parameter ε becomes smaller, the smoothed feature aggregating K_r-hops neighbors for the block b_r will be more likely to present oversmoothing because the distance of smoothed feature via aggregating K_r-hops neighbors information within block structure b_r will become larger. Oversmoothing phenomenon will introduce interclass information leading to a large difference in feature characteristics. Finally, the optimal K_r-hops neighbors for different imbalanced regions represented by different block structures b_r can be obtained via (15). The final optimal feature representation for imbalanced class regions can be obtained by using the matrix form

$$\hat{\boldsymbol{X}} = \|_{r=1}^{C} \hat{\boldsymbol{S}}_{b_r}^{K_r} \cdot \boldsymbol{X}_{b_r} \tag{18}$$

and $\|_{r=1}^{C}$ denotes the feature concatenation operation from the C block. The classification results can be obtained by

$$Z_{\text{CIGCS}} = \text{softmax}\left(\hat{X}\Theta\right). \tag{19}$$

The classifier is inherited from the SGC model followed by a linear logistic regression classifier. The completed procedure for CIGCS model is elaborately described in Algorithm 1. Besides, the theoretical property of adaptive K_r-hops neighbors should be further analyzed to ensure its upper bound instead of approximating infinite hops neighbors. The specific theory is introduced as follows, and its proof can be found in the Appendix.

Algorithm 1 CIGCS Model for HSI Classification

Input: Given the HSI data matrix $\boldsymbol{X}=[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{N\times d}$ and the spatial correlation matrix $\mathbf{L}=[l_1, \ldots, l_N] \in \mathbb{R}^{N\times 2}$.

1) Obtain the block-diagonal graph structure $\boldsymbol{S}$ with improved intraclass connectivity and suppressed interclass connectivity to describe imbalanced region distribution by using label guidance and Laplacian regularizer via Eq. 13.
2) Obtain the optimal K_r-hops neighbors to achieve the optimal feature smoothing for each block b_r via Eq. 15.
3) Obtain the optimal smoothed feature and perform the classification using a linear logistic regression classifier via Eq. 19.

Output: Classification results of the testing samples.

Theorem 1: Given the block-diagonal graph structure $\boldsymbol{S}_{b_r}^{K_r}$ and feature smoothing $\hat{\boldsymbol{X}}_{b_r} = \hat{\boldsymbol{S}}_{b_r}^{K_r} \cdot \boldsymbol{X}_{b_r}$ with $\hat{\boldsymbol{S}}_{b_r}^{K_r} = \tilde{\boldsymbol{D}}^{-1}\tilde{\boldsymbol{S}}_{b_r}^{K_r}$, we can derive

$$K_r(b_r, \varepsilon) \le \log_{\lambda_2}\left(\varepsilon\sqrt{\frac{\tilde{d}_i}{2M_{b_r} + N_{b_r}}}\right) \tag{20}$$

where $\lambda_2 \le 1$ is the second largest eigenvalue of $\hat{\boldsymbol{S}}_{b_r}^{K_r}$, $\tilde{d}_i$ denotes the degree of node v_i, i.e., $\tilde{d}_i = d_i + 1$, and M_{b_r} and N_{b_r} denote the number of edges and nodes, respectively.

From problem (21), the adaptive optimal K_r-hops neighbors are positively correlated with the scale of the M_{b_r} and N_{b_r}, the sparsity of the graph (small λ_2 means strong connection and low sparsity, and vice versa), and negatively correlated with the degree of node v_i.

IV. Optimization and Algorithm Analysis

In this section, the main work is to optimize and analyze the CIGCS model. First, the block-diagonal structure of the CIGCS model is optimized by an efficient iterative algorithm. Next, the analysis and complexity of the CIGCS model are introduced.

A. Optimization for Block-Diagonal Model

The problem (13) can be equivalent to the following problem:

$$\min_{\boldsymbol{s}_i^{\mathrm{T}}\mathbf{1}=1, 0\le \boldsymbol{s}_i\le 1} \| \mathrm{S} - \boldsymbol{Q} \|_F^2 + 2\varphi \sum_{i=0}^{C} \Lambda_i(\boldsymbol{L}_S) + \lambda\left(\sum_{i,j=1}^{N}\left(\| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 + \beta \| \boldsymbol{l}_i - \boldsymbol{l}_j \|_2^2\right)s_{ij}\right). \tag{21}$$

According to Ky Fan's theorem [47], we can obtain the following equation:

$$\sum_{i=0}^{C} \Lambda_i(\boldsymbol{L}_s) = \min_{\boldsymbol{F}\in\mathbb{R}^{N\times C}, \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}=\boldsymbol{I}} Tr\left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{L}_S\boldsymbol{F}\right) \tag{22}$$

where $\boldsymbol{F}$ denotes the C eigenvectors of corresponding to the C smallest eigenvalues. Therefore, the problem (21) is further equivalent to the following problem:

$$\begin{aligned}\min\ & \| \mathrm{S} - \boldsymbol{Q} \|_F^2 + 2\varphi Tr\left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{L}_S\boldsymbol{F}\right) + \lambda\left(\sum_{i,j=1}^{N}\left(\| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 + \beta \| \boldsymbol{l}_i - \boldsymbol{l}_j \|_2^2\right)\boldsymbol{s}_{ij}\right) \\ s.t.\ & \boldsymbol{s}_i^{\mathrm{T}}\mathbf{1} = 1,\ 0 \le \boldsymbol{s}_i \le 1, \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{I}.\end{aligned} \tag{23}$$

Next, fix $\boldsymbol{Q}$ and $\boldsymbol{F}$ and update $\boldsymbol{S}$. The problem (23) can be rewritten as the vector form

$$\min_{\boldsymbol{s}_i^{\mathrm{T}}\mathbf{1}=\mathbf{1},\ 0\le \boldsymbol{s}_i} \sum_{i,j}\left(s_{ij} - q_{ij}\right)^2 + \varphi\sum_{i,j=1}^{N} \| \boldsymbol{f}_i - \boldsymbol{f}_j \|_2^2 s_{ij} + \lambda\left(\sum_{i,j=1}^{N}\left(\| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 + \beta \| \boldsymbol{l}_i - \boldsymbol{l}_j \|_2^2\right)s_{ij}\right). \tag{24}$$

Note that problem (24) is independent for different i, and thus, problem (24) can be solved separately for each i

$$\min_{\sum_j s_{ij}=1, s_{ij}>0} \sum_j \left(s_{ij} - q_{ij}\right)^2 + \lambda\sum_j d_{ij}s_{ij} \tag{25}$$

where $d_{ij} = \| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2 + \beta \| \boldsymbol{l}_i - \boldsymbol{l}_j \|_2^2 + (\varphi/\lambda) \| \boldsymbol{f}_i - \boldsymbol{f}_j \|_2^2$ and $\boldsymbol{d}_i$ denotes a vector with the jth element equal to d_{ij}. The problem (25) can be written as

$$\min_{\boldsymbol{s}_i^{\mathrm{T}}\mathbf{1}=1, \boldsymbol{s}_i\ge 0} \| \boldsymbol{s}_i - \left(\boldsymbol{q}_i - \frac{\lambda}{2}\boldsymbol{d}_i\right) \|_F^2. \tag{26}$$

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This problem can be solved with a closed-form solution by an efficient iterative algorithm in [48]. Update the label connectivity $\boldsymbol{Q}$ via the updated $\boldsymbol{S}$ in (12), and update $\boldsymbol{F}$ via the eigenvalue decomposition of Laplacian matrix $\boldsymbol{L}_s$.

B. Convergence Analysis and Complexity Analysis

For the CIGCS model, convergence arises from the construction of the block-diagonal structure, First, $\boldsymbol{S}$ can be solved with a closed form using an efficient iterative algorithm, which has been proven to converge. The description of the class-imbalanced region with $\boldsymbol{S}$ will further construct the label connectivity and eigenvalue decomposition of Laplacian matrix $\boldsymbol{L}_s$. Then, the convergence of the logistic regression can be guaranteed by the gradient descent algorithm. Overall, the convergence of the CIGCS model can be guaranteed. The computational complexity arises from the solution of the similarity matrix $\boldsymbol{S}$, label connectivity $\boldsymbol{Q}$, and the eigenvalue decomposition of Laplacian matrix $\boldsymbol{L}_s$. The complexity of $\boldsymbol{S}$ is about $O(N)$ because the solution of $\boldsymbol{S}$ can be obtained independently for every i. The complexity of $\boldsymbol{Q}$ is from the matrix multiplication of the label distribution about $O(N^2)$. The complexity of the eigenvalue decomposition of Laplacian matrix $\boldsymbol{L}_s$ is about $O(N^2)$. In summary, the computational complexity of the CIGCS model is about $O(N^2)$.

V. Experimental Results and Analysis

In this section, in order to fully exploit the potential of the CIGCS method, some comprehensive experiments on three real HSI datasets illustrate the promising classification performance of the proposed CIGCS model in the HSI classification. Moreover, the deep learning-based methods, including ASSMN methods, and the GCN-based methods, including GCN, SGC, DAS-GCN, CEGCN, MDGCN, GCGCN, and SSAPGCN methods, are compared with the CIGCS.

A. Dataset Description

The experimental results are obtained based on three typical HSI datasets, Indian Pines, Pavia University (PaviaU), and Houston University 2013 [49], [50].

1) The Indian Pines image, captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, had a size of 145×145 with 220 bands. Moreover, the spatial and spectral resolutions of this image were 20 m and 10 nm, respectively. This scene contained 16 classes covering an agricultural field. The false-color image and its corresponding ground truth are given in Fig. 3(a) and (b).

2) The Houston University (Houston) dataset was captured by the Compact Airborne Spectrographic Imager (CASI) over the University of Houston campus and the neighboring urban area in June 2013. The Houston image contains 349 $\times$ 1905 pixels with 144 spectral bands. In addition, the spatial resolution was 2.5 m. The false color and the corresponding ground truth of Houston are shown in Fig. 4(a) and (b).

3) The HyRANK dataset, namely Dioni, was from the Hyperion sensor (EO-1, USGS). The Dioni dataset contains 250 $\times$ 1376 pixels with 176 spectral bands, which contains 12 land-cover types in the dataset. The false-color and corresponding ground-truth images of Dioni are shown in Fig. 5(a) and (b).

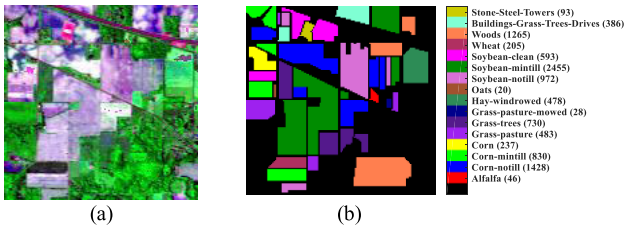


Fig. 3. Indian Pines data: (a) false-color image and (b) ground-truth image (bands 20, 69, and 178 are red, green, and blue, respectively).

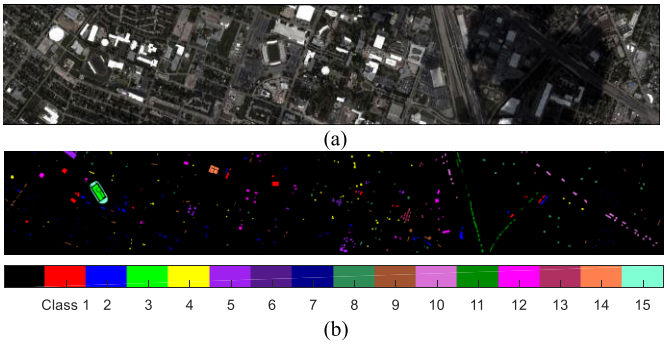


Fig. 4. Houston data: (a) false-color image and (b) ground-truth image (bands 45, 38, and 32 are red, green, and blue, respectively).

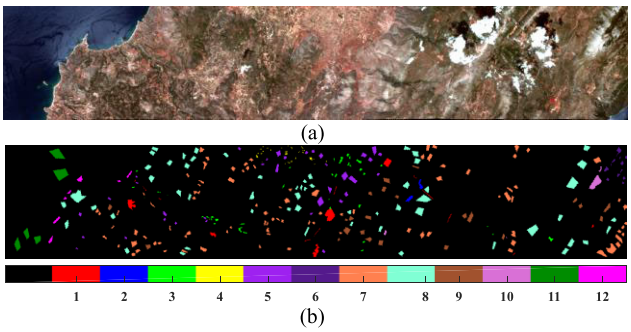


Fig. 5. HyRank data with Dioni: (a) false-color image (bands 23, 11, and 7 are red, green, and blue, respectively) and (b) ground-truth image.

B. Experimental Settings

In our experiments, the proposed CIGCS method is implemented on PyTorch with Adam optimizer. For three datasets including Indian Pines, Houston 2013, and Dioni, one training sample is randomly selected in each class for training, five samples are randomly selected in each class for validation to tune the hyperparameters, and the remainder of the samples are selected as the testing set. The numbers of the training and testing samples in the experiments are shown in detail in Table I. The learning rate and the training epochs are set to 0.05 and 200 for three datasets, respectively. To alleviate the effects of random initialization on the classification performance, all the experiments are repeatedly implemented ten times, and the average results are recorded. To quantitatively evaluate the performance of each model, the per-class accuracy, the overall accuracy (OA), the average accuracy (AA), and the

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TABLE I

DETAILED NUMBER OF TRAINING AND TESTING SAMPLES ON INDIAN PINES, HOUSTON, AND DIONI DATASETS

Indian Pines					Houston					Dioni				
Class	Name	Train	Validation	Test	Class	Name	Train	Validation	Test	Class	Name	Train	Validation	Test
1	Alfalfa	1	5	40	1	Healthy grass	1	5	1245	1	Dense Urban Fabric	1	5	1256
2	Corn-notill	1	5	1422	2	Stressed grass	1	5	1248	2	Mineral Extraction Sites	1	5	198
3	Corn-mintill	1	5	824	3	Synthetic grass	1	5	691	3	Non Irrigated Arable Land	1	5	608
4	Corn	1	5	231	4	Tree	1	5	1238	4	Fruit Trees	1	5	144
5	Grass-pasture	1	5	477	5	Soil	1	5	1236	5	Olive Groves	1	5	1762
6	Grass-trees	1	5	724	6	Water	1	5	319	6	Coniferous Forest	1	5	355
7	Grass-pasture-mowed	1	5	22	7	Residential	1	5	1262	7	Dense Sclerophyllous Vegetation	1	5	5029
8	Hay-windrowed	1	5	472	8	Commercial	1	5	1238	8	Sparce Sclerophyllour Vegetation	1	5	6368
9	Oats	1	5	14	9	Road	1	5	1246	9	Sparcely Vegetated Areas	1	5	1748
10	Soybean-notill	1	5	966	10	Highway	1	5	1221	10	Rock and Sand	1	5	486
11	Soybean-mintill	1	5	2449	11	Railway	1	5	1229	11	Water	1	5	1606
12	Soybean-clean	1	5	587	12	Parking lot 1	1	5	1227	12	Coastal Water	1	5	392
13	Wheat	1	5	199	13	Parking lot 2	1	5	463		Total	12	60	19952
14	Woods	1	5	1259	14	Tennis court	1	5	422					
15	Buildings-Grass-Trees-Drives	1	5	380	15	Running track	1	5	654					
16	Stone-Steel-Towers	1	5	87		Total	15	75	14939					
	Total	16	80	10603										

TABLE II

CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class	ASSMN	GCN	DAS-GCN	SGC	MDGCN	GCGCN	SSAPGCN	CEGCN	WFCG	HybridFormer	MARP	CIGCS
1	77.11 (25.06)	87.81 (7.07)	42.01 (23.83)	75.46 (33.84)	91.48 (0.91)	93.71 (3.83)	31.48 (6.35)	68.86 (21.51)	83.65 (17.56)	93.57 (17.01)	**99.44** (1.10)	98.00 (0.79)
2	17.06 (6.40)	21.27 (17.50)	51.86 (21.16)	21.15 (7.51)	19.14 (15.62)	23.84 (12.01)	31.82 (3.68)	22.96 (16.47)	39.51 (13.07)	**51.73** (13.39)	44.20 (21.06)	50.83 (17.08)
3	21.54 (16.04)	26.34 (20.63)	36.61 (22.67)	38.16 (22.28)	26.22 (17.88)	30.80 (8.59)	0 (0)	33.99 (20.49)	50.40 (12.95)	36.16 (7.45)	**60.26** (11.04)	52.91 (17.71)
4	34.58 (7.68)	19.99 (12.25)	28.66 (8.95)	86.89 (21.92)	43.86 (22.16)	40.39 (10.86)	16.88 (2.56)	30.86 (14.97)	62.33 (12.73)	87.81 (8.15)	76.14 (32.62)	**89.70** (15.76)
5	26.18 (12.47)	41.42 (31.09)	91.16 (12.17)	51.52 (22.98)	46.75 (25.21)	16.94 (24.37)	5.64 (0.93)	46.12 (23.85)	74.27 (15.74)	70.13 (22.14)	**79.38** (10.24)	65.70 (8.31)
6	22.92 (10.33)	36.92 (9.49)	**94.75** (5.18)	45.03 (40.73)	30.43 (16.52)	46.25 (12.88)	19.11 (2.78)	84.65 (15.90)	81.48 (13.32)	94.41 (1.68)	88.96 (11.33)	71.05 (13.08)
7	92.96 (14.34)	69.14 (25.19)	54.58 (33.94)	56.92 (29.11)	97.53 (6.05)	97.53 (1.91)	35.19 (6.93)	86.54 (13.57)	48.47 (22.02)	96.75 (8.59)	**100** (0)	**100** (0)
8	70.63 (30.46)	25.51 (11.29)	97.59 (2.74)	86.81 (22.91)	91.13 (11.56)	78.51 (26.25)	73.60 (1.66)	50.72 (20.68)	95.91 (6.14)	85.77 (11.17)	94.89 (3.50)	**100** (0)
9	65.26 (20.03)	54.39 (30.08)	43.97 (29.12)	85.71 (48.30)	100 (0)	88.60 (17.74)	0 (0)	77.22 (25.02)	44.51 (34.35)	53.06 (20.14)	84.13 (34.59)	**100** (0)
10	26.79 (21.77)	26.31 (20.17)	42.05 (22.17)	56.84 (27.68)	35.99 (23.19)	47.96 (17.59)	9.46 (1.11)	53.18 (13.73)	63.74 (8.16)	54.27 (6.81)	58.72 (9.84)	**82.67** (24.81)
11	41.15 (26.41)	37.05 (17.83)	**69.74** (18.09)	66.12 (22.98)	32.76 (15.54)	57.05 (20.82)	18.70 (2.78)	42.96 (23.28)	54.73 (13.15)	58.71 (12.59)	57.76 (14.94)	39.01 (35.06)
12	15.15 (9.01)	20.24 (8.69)	31.78 (8.62)	27.87 (10.56)	29.14 (9.65)	32.80 (8.14)	24.30 (1.40)	15.83 (16.97)	48.13 (9.76)	35.09 (10.26)	30.27 (11.21)	**72.06** (22.98)
13	48.43 (17.14)	80.80 (14.27)	83.69 (10.32)	99.01 (0)	93.87 (6.80)	97.63 (3.18)	85.70 (3.84)	94.7 (13.31)	79.52 (10.75)	93.89 (7.33)	98.21 (4.82)	**99.19** (0.34)
14	66.10 (23.93)	57.99 (18.00)	96.31 (4.76)	74.47 (33.02)	64.98 (29.35)	87.37 (21.64)	18.64 (2.03)	70.6 (26.35)	88.63 (5.94)	86.63 (7.40)	74.75 (17.21)	**99.90** (0.03)
15	43.56 (22.93)	13.29 (6.51)	49.27 (13.77)	50.32 (24.88)	34.16 (16.91)	21.91 (15.19)	10.48 (0.78)	36.35 (23.81)	57.59 (18.35)	73.68 (24.44)	**86.26** (10.65)	74.05 (11.61)
16	82.94 (18.16)	86.77 (6.50)	49.14 (32.12)	96.48 (1.62)	94.93 (5.70)	96.92 (4.88)	76.45 (1.78)	92.42 (6.60)	81.81 (26.01)	91.62 (15.91)	97.19 (6.73)	**99.08** (1.38)
OA	36.59 (6.27)	34.59 (6.20)	52.29 (5.23)	51.12 (4.97)	40.26 (6.22)	49.90 (3.03)	21.99 (0.57)	46.97 (7.45)	63.51 (3.95)	64.32 (4.11)	64.55 (2.84)	**66.36** (5.85)
AA	47.02 (3.48)	44.08 (6.77)	60.20 (5.96)	60.49 (4.60)	58.27 (4.64)	59.89 (3.45)	28.59 (0.50)	56.75 (5.31)	66.47 (2.79)	72.71 (5.08)	76.91 (4.89)	**80.88** (2.39)
Kappa	30.05 (5.80)	27.18 (6.59)	47.68 (5.43)	44.56 (5.34)	33.81 (6.38)	43.24 (3.23)	13.90 (0.39)	40.59 (7.19)	58.92 (4.02)	60.16 (4.53)	60.54 (3.26)	**62.75** (5.83)

Kappa coefficient (KC) are adopted as indicators. The closer the metrics are to 1, the better the classification results.

C. Experimental Results

1) Table II shows the classification results of different classification methods on Indian Pines data. Compared to the CNN-based method, namely ASSMN method, most GCN-based methods achieve better classification results. This is because the irregular graph convolution, as opposed to regular convolution operation, is more suitable for capturing complex spatial–spectral information within irregular regions, especially when there are limited training samples. Among

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TABLE III
CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE HOUSTON 2013 DATASET

Class	ASSMN	GCN	DAS-GCN	SGC	MDGCN	GCGCN	SSAPGCN	CEGCN	WFCG	HybridFormer	MARP	CIGCS
1	44.94 (28.73)	**94.70** (7.10)	70.89 (13.40)	63.43 (34.03)	81.80 (3.48)	61.67 (36.80)	77.34 (15.90)	42.29 (26.52)	68.04 (10.56)	82.31 (6.68)	70.32 (6.87)	63.30 (33.99)
2	17.96 (12.66)	44.53 (19.54)	76.22 (7.17)	40.39 (20.99)	61.48 (28.15)	**80.30** (8.11)	50.00 (28.75)	21.71 (12.80)	40.59 (10.36)	47.96 (25.24)	52.72 (21.47)	74.94 (11.95)
3	35.71 (25.69)	88.88 (25.81)	88.30 (7.68)	83.89 (33.75)	80.32 (35.31)	**100** (0)	99.40 (0.38)	99.98 (0.04)	92.83 (5.12)	96.38 (3.01)	62.08 (16.68)	99.41 (0.64)
4	27.34 (19.44)	71.41 (25.03)	62.96 (20.91)	77.07 (25.36)	72.49 (33.90)	64.33 (29.72)	80.21 (28.98)	71.95 (13.91)	69.56 (8.85)	84.31 (15.59)	63.61 (15.53)	**90.71** (6.46)
5	35.45 (22.51)	61.80 (16.61)	71.18 (12.82)	61.35 (23.32)	94.13 (4.21)	90.89 (13.47)	63.47 (24.01)	80.88 (14.32)	87.47 (7.05)	86.61 (8.23)	86.61 (18.59)	**95.21** (3.33)
6	16.82 ((8.28)	66.11 (30.36)	81.41 (22.56)	83.47 (0.59)	55.86 (25.40)	84.73 (0.26)	66.11 (31.58)	61.47 (32.07)	88.03 (5.08)	77.83 (4.53)	85.58 (4.43)	**86.49** (3.83)
7	10.43 (6.69)	21.46 (10.30)	46.99 (23.86)	18.18 (17.76)	**59.24** (19.82)	33.40 (18.55)	44.08 (17.89)	22.93 (14.37)	55.42 (9.21)	45.55 (25.68)	58.83 (8.91)	37.23 (19.19)
8	11.01 (10.03)	15.66 (7.78)	**62.73** (21.31)	10.48 (10.04)	10.20 (7.43)	14.32 (6.76)	12.70 (13.08)	14.55 (10.50)	24.70 (10.81)	7.66 (7.81)	33.60 (3.59)	15.78 (9.37)
9	20.23 (11.88)	57.85 (28.49)	33.49 (15.93)	55.34 (15.52)	46.37 (30.35)	48.71 (24.14)	59.35 (23.86)	30.55 (27.53)	40.17 (13.25)	**68.86** (11.98)	54.17 (29.96)	66.69 (10.35)
10	12.98 (11.47)	24.94 (5.66)	32.50 (7.88)	25.08 (10.27)	26.66 (5.26)	26.82 (8.32)	27.14 (4.61)	32.43 (9.78)	36.32 (4.35)	31.40 (3.43)	**63.14** (2.26)	49.76 (16.53)
11	18.27 (18.56)	32.91 (20.48)	26.39 (6.40)	56.96 (22.79)	52.23 (26.54)	70.54 (2.82)	38.14 (24.14)	57.26 (16.68)	52.09 (13.06)	44.06 (16.87)	61.88 (15.88)	**78.26** (5.38)
12	19.59 (13.58)	21.36 (7.80)	35.57 (14.94)	12.16 (7.18)	35.95 (17.27)	20.63 (8.42)	23.51 (13.70)	23.03 (17.85)	**53.13** (8.88)	32.26 (20.41)	42.71 (7.95)	22.93 (10.08)
13	26.11 (19.63)	15.11 (7.83)	37.32 (10.02)	15.67 (13.59)	41.15 (21.58)	35.01 (21.47)	14.66 (11.42)	19.46 (22.63)	51.24 (7.72)	67.47 (28.79)	**78.29** (18.48)	30.69 (8.52)
14	34.99 (13.23)	66.49 (26.94)	97.19 (1.85)	97.54 (1.11)	**100** (0)	**100** (0)	59.30 (18.14)	95.08 (12.45)	96.96 (3.06)	91.26 (8.09)	99.88 (0.17)	97.42 (5.14)
15	19.80 (5.69)	74.36 (19.76)	93.08 (8.74)	95.53 (3.37)	84.10 (19.56)	**99.19** (0.15)	97.24 (2.87)	89.77 (28.25)	92.80 (7.92)	95.98 (7.70)	84.02 (2.28)	97.74 (2.06)
OA	22.78 (4.25)	48.20 (4.61)	51.99 (3.01)	49.28 (3.36)	56.84 (4.50)	57.17 (1.85)	51.95 (5.33)	46.12 (2.42)	61.51 (2.73)	60.08 (2.81)	62.15 (2.56)	**63.76** (3.22)
AA	23.44 (3.93)	50.50 (4.41)	61.08 (3.64)	55.10 (2.94)	60.13 (4.72)	62.04 (2.12)	54.18 (6.03)	50.89 (2.62)	63.29 (2.13)	63.99 (3.09)	65.85 (1.76)	**67.10** (2.59)
Kappa	17.00 (4.52)	44.31 (4.88)	48.18 (3.23)	45.01 (3.58)	53.73 (4.59)	54.00 (2.04)	48.15 (5.75)	41.80 (2.54)	57.63 (2.85)	56.95 (2.97)	59.35 (2.39)	**60.94** (3.46)

the GCN-based methods, the SGC method, which decouples feature propagation and feature transformation, serves as the baseline for the CIGCS method. In terms of classification performance, the SGC method demonstrates a slight superiority over other GCN-based methods, including GCN, MDGCN, and CEGCN. This finding demonstrates that the decoupled methods, such as SGC method, exhibit competitive capability in feature extraction compared to end-to-end GCN-based methods, especially when dealing with limited training samples. However, the SGC method overlooks the class-imbalanced information in feature extraction, resulting in limited classification performance. To address this problem, the CIGCS method was proposed, which captures adaptive neighbors' information for different class distribution and considers the class-imbalanced problem in feature extraction. As a result, the CIGCS method outperforms other GCN-based classification methods.

2) Table III shows the classification results of different classification methods on Houston data. Although the irregular graph convolution can flexibly aggregate neighboring information for feature smoothing, the classification performance of the GCN, MDGCN, GCGCN, and CEGCN methods is inferior to that of the proposed CIGCS method. The MDGCN method leverages diversified spectral–spatial correlations at multiple scales capturing multiscale neighboring information through dynamically updated graph convolution processes. It can partially mitigate the influence of imbalanced class regions in feature extraction due to the aggregation of multiscale neighboring information. However, the CIGCS method achieves superior classification performance compared to the MDGCN method because it focuses more on flexible aggregation of neighboring information through a compact graph structure and adaptive neighboring information extraction for imbalanced class regions. Consequently, the CIGCS exhibits a 7% improvement in OA, KC, and AA indicators compared to the MDGCN method.

3) Table IV shows the classification results of different classification methods on Dioni data. Overall, the CIGCS method outperforms other GCN-based methods in classification performance. In comparison to the SSAPGCN method, which employs a spatial–spectral local adaptive probability graph structure, the CIGCS method utilizes a block-diagonal structure to describe imbalanced class regions from both local and global perspectives. This approach enhances the intraclass connectivity and suppresses the interclass connectivity within the graph structure. Consequently, the CIGCS method achieves better OA than the SSAPGCN method. Besides, compared to the GCGCN method, which constructs local and global graphs based on pixel- and superpixel-level views, the CIGCS method achieves superior classification performance

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TABLE IV
CLASSIFICATION RESULTS (STD) (%) OF DIFFERENT METHODS ON THE DIONI DATASET

Class	ASSMN	GCN	DAS-GCN	SGC	MDGCN	GCGCN	SSAPGCN	CEGCN	WFCG	HybridFormer	MARP	CIGCS
1	20.95 (14.96)	29.06 (20.77)	33.95 (12.77)	52.02 (20.34)	69.42 (27.41)	60.40 (14.52)	40.11 (23.15)	29.38 (15.77)	58.68 (21.26)	64.00 (12.52)	47.00 (30.33)	**70.79** (13.49)
2	32.81 (16.94)	49.26 (17.94)	23.65 (6.37)	65.40 (28.13)	48.44 (5.32)	74.71 (12.68)	35.91 (15.11)	50.20 (20.77)	74.79 (8.24)	91.85 (7.71)	**95.39** (1.82)	92.52 (3.58)
3	31.98 (25.16)	38.42 (23.22)	38.20 (26.24)	57.96 (25.31)	44.45 (32.91)	44.59 (35.70)	47.52 (21.55)	24.62 (18.65)	50.88 (10.15)	68.52 (17.28)	71.18 (4.76)	**73.78** (13.30)
4	51.48 (18.10)	75.91 (22.13)	17.46 (8.48)	88.04 (9.61)	59.62 (35.91)	77.74 (28.38)	68.86 (12.13)	70.34 (31.16)	40.42 (12.91)	87.06 (11.32)	63.89 (28.25)	**94.65** (5.45)
5	23.09 (17.25)	46.80 (18.41)	43.16 (13.48)	55.01 (20.91)	26.29 (11.91)	46.42 (15.78)	32.87 (15.54)	31.73 (23.33)	53.69 (20.92)	49.08 (18.03)	**80.48** (5.06)	47.90 (13.73)
6	40.50 (19.13)	97.92 (2.69)	31.98 (6.86)	99.97 (0.09)	100 (0)	100 (0)	98.86 (1.58)	84.79 (25.26)	76.46 (5.02)	98.10 (0.42)	100 (0)	**100** (0)
7	32.99 (12.68)	51.12 (26.82)	**69.56** (9.06)	56.80 (17.55)	40.64 (18.82)	60.56 (23.08)	53.21 (26.28)	33.06 (22.62)	74.33 (0.56)	83.49 (6.40)	65.93 (19.39)	65.31 (18.59)
8	31.94 (24.21)	27.55 (17.18)	62.94 (11.25)	51.42 (15.11)	31.09 (21.86)	63.18 (17.64)	36.79 (22.57)	30.86 (9.53)	70.32 (8.25)	55.32 (14.65)	70.64 (23.72)	**74.02** (14.01)
9	27.22 (13.93)	29.23 (14.95)	36.12 (8.51)	36.20 (12.03)	36.66 (25.28)	31.28 (22.43)	30.56 (14.51)	37.27 (14.00)	51.23 (2.13)	57.68 (12.08)	40.86 (31.03)	**64.24** (14.06)
10	38.96 (26.50)	83.57 (12.16)	30.90 (7.13)	**91.02** (3.16)	66.43 (32.87)	76.68 (32.77)	84.20 (11.91)	83.03 (24.82)	73.39 (8.07)	90.09 (0.43)	66.23 (26.05)	90.95 (2.89)
11	90.81 (28.81)	51.29 (19.50)	76.81 (14.10)	65.02 (17.97)	**96.37** (4.00)	72.31 (21.16)	60.83 (26.20)	67.53 (23.55)	97.22 (2.12)	80.57 (23.44)	100 (0)	90.97 (18.60)
12	66.95 (21.58)	58.89 (23.16)	57.36 (22.64)	47.48 (12.44)	77.25 (19.64)	45.93 (17.50)	66.02 (25.96)	61.15 (23.96)	79.18 (7.17)	87.56 (12.77)	**90.75** (4.94)	80.97 (12.36)
OA	36.23 (8.37)	41.51 (9.44)	47.63 (5.35)	55.23 (8.77)	45.05 (5.26)	59.12 (5.00)	45.59 (12.38)	38.06 (7.94)	68.09 (0.59)	67.89 (3.79)	69.61 (3.69)	**71.16** (2.87)
AA	40.81 (7.67)	53.25 (4.97)	43.51 (4.69)	63.86 (6.38)	58.05 (3.71)	62.82 (4.22)	54.65 (5.83)	50.33 (6.63)	67.13 (2.92)	76.11 (2.94)	74.36 (5.76)	**78.84** (2.07)
Kappa	26.54 (8.12)	33.17 (10.09)	39.93 (5.58)	47.5 9(9.62)	37.11 (5.63)	51.68 (5.10)	37.30 (12.83)	29.30 (7.73)	61.39 (1.57)	61.61 (4.34)	62.62 (4.77)	**64.78** (3.50)

by aggregating adaptive neighboring information according to the natural class-imbalanced property instead of capturing the fixed K-hops neighbors for information propagation such as the GCGCN method. Next, the classification performance of the CIGCS method is better than that of the WFCG, HybridFormer, and MARP methods with an improvement of about 2% in OA, AA, and KC indicators. This phenomenon demonstrates that capturing adaptive hop neighbors for each semantic class is beneficial to achieve effective feature smoothing for downstream classification task due to the consistence of imbalanced class distributions. Figs. 6–8 demonstrate that the classification map generated by the CIGCS method is closer to the ground truth compared to those of other GCN-based classification methods. This phenomenon is also proving that the classification performance of the proposed CIGCS method is beneficial to the application of downstream task.

D. Analysis of Training Information

To further demonstrate the superiority of the CIGCS method over other classification methods, the related experiments were conducted on three datasets. Fig. 9 shows the OA curve with different numbers of training samples. From Fig. 9, it can be observed that the CIGCS method consistently outperforms other classification methods across different training sample sizes. Notably, most classification methods perform poorly when there is only one training sample, as they struggle to learn discriminative features due to spectral variations. In contrast, the CIGCS method can learn more discriminative features using only one training sample by flexibly aggregating effective feature information from adaptive neighbors within different semantic class regions achieving optimal feature smoothing. With an increase in training samples, most classification methods exhibit improved performance as the capability of feature extraction operation improves. As a result, the classification performance gap between most methods does not appear significant.

E. Ablation Study

The proposed CIGCS method mainly consists of two critical parts, namely, semantic block-diagonal structure and class-imbalanced graph convolution, to enhance the classification performance. To demonstrate the usefulness of these two critical parts, the related experiments were conducted on three datasets, as shown in Table V. The SGC method serves as the base model, and the CIGCS model represents its improved version. The related experiments compare the SGC model with added parts: SGCB method, which incorporates the improvement from block-diagonal structure, and the SGCIM method, which includes the enhancement from class-imbalanced graph convolution. According to Table V, the classification performance of the CIGCSM method consistently outperforms the other methods. Notably, the SGCB method exhibits better improvement compared to the SGC model compared to the SGCIM method. This is primarily due to the ability of the SGCB method to describe intraclass distributions and suppress interclass connections using the block-diagonal structure. Hence, constructing a block-diagonal structure to describe imbalanced class regions is crucial for the CIGCS method. Both the SGCB method and the SGCIM method show promising improvements over the SGC model, highlighting

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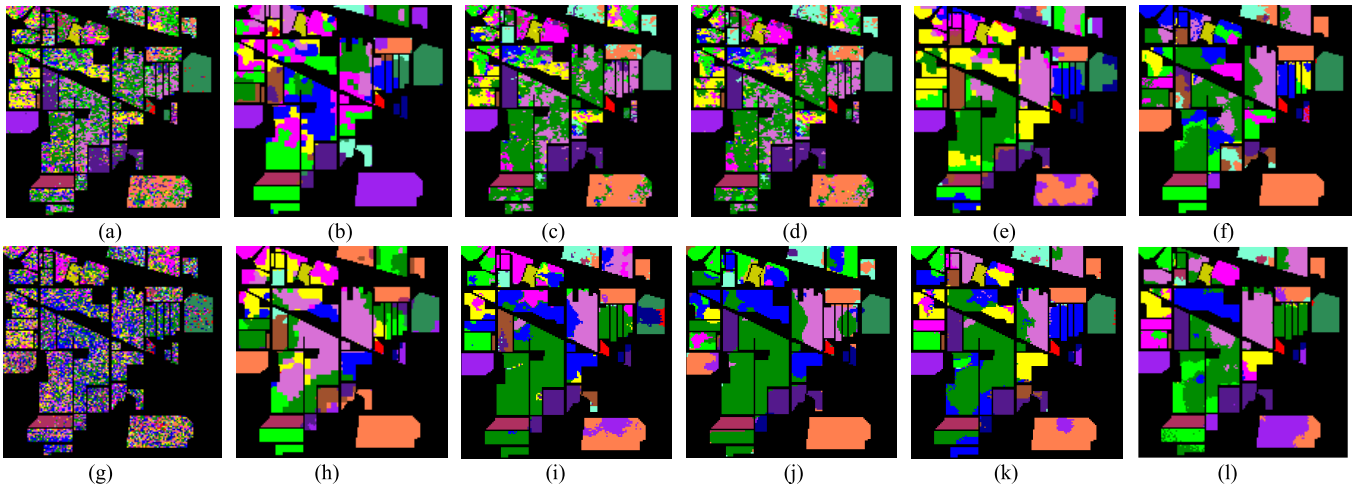


Fig. 6. Classification maps of different methods on the Indian Pines dataset. (a) ASSMN. (b) GCN. (c) DAS-GCN. (d) SGC. (e) MDGCN. (f) GCGCN. (g) SSAPGCN. (h) CEGCN. (i) WFCG. (j) HybridFormer. (k) MARP. (l) CIGCS.

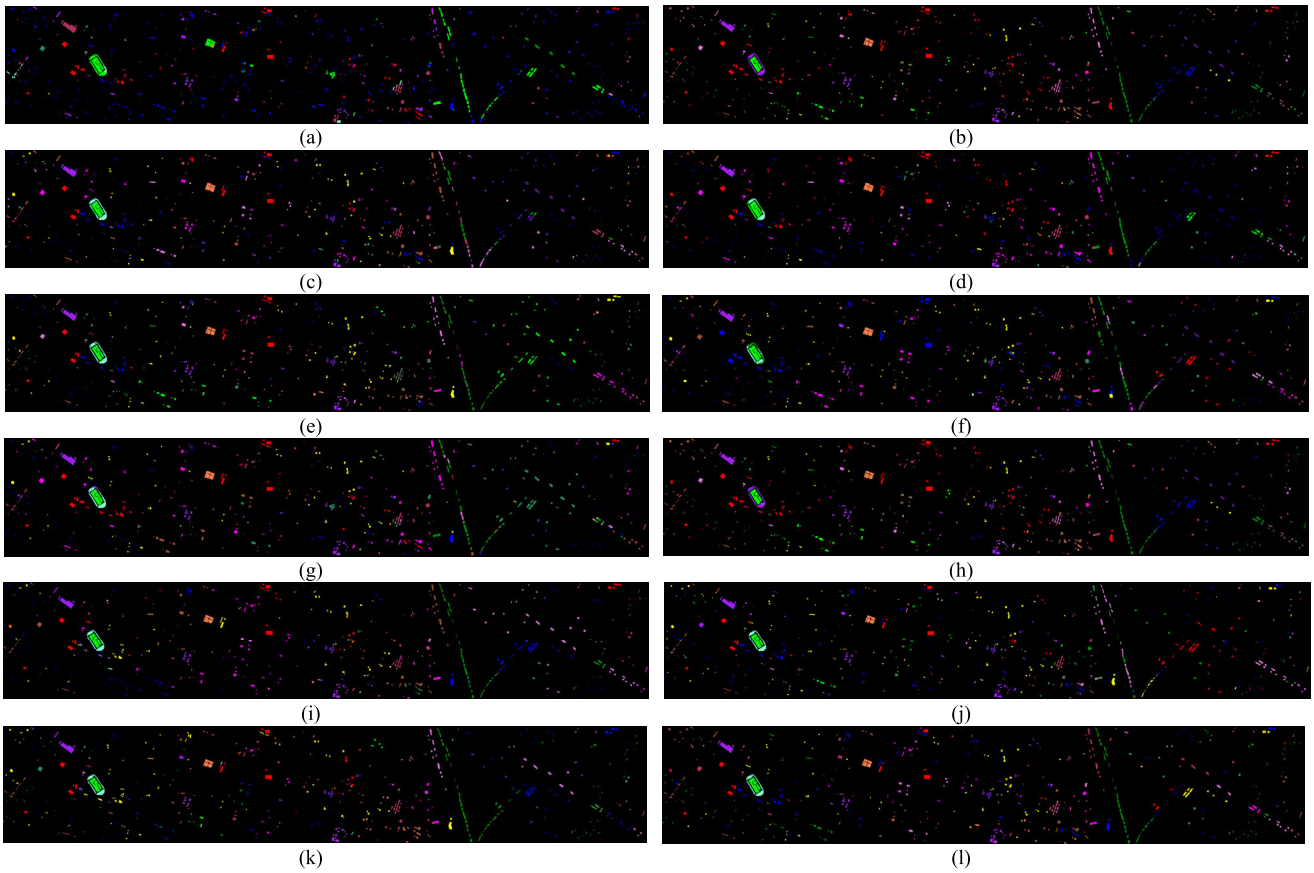


Fig. 7. Classification maps of different methods on the Houston 2013 dataset. (a) ASSMN. (b) GCN. (c) DAS-GCN. (d) SGC. (e) MDGCN. (f) GCGCN. (g) SSAPGCN. (h) CEGCN. (i) WFCG. (j) HybridFormer. (k) MARP. (l) CIGCS.

the effectiveness of the semantic block-diagonal structure and class-imbalanced graph convolution in the CIGCS method.

F. Analysis of Tradeoff Parameters

The proposed CIGCS method involves four adjustable parameters, including λ, φ, β, K_r, and ε for obtaining promising classification performance on three datasets. First, the parameter β balances spectral information and spatial information, and it can be set 30 empirically according to [27]. Through the above analysis, parameters λ and φ need to be adjusted carefully in order to downstream feature smoothing. Thus, experiments were conducted on three datasets with varying values of parameters λ and φ that are conducted to obtain the optimal parameter value, which can be observed in Fig. 10. The parameter λ controls the spatial–spectral information for obtaining local graph structure. Also, the parameter φ denotes the degree of the ideal block-diagonal

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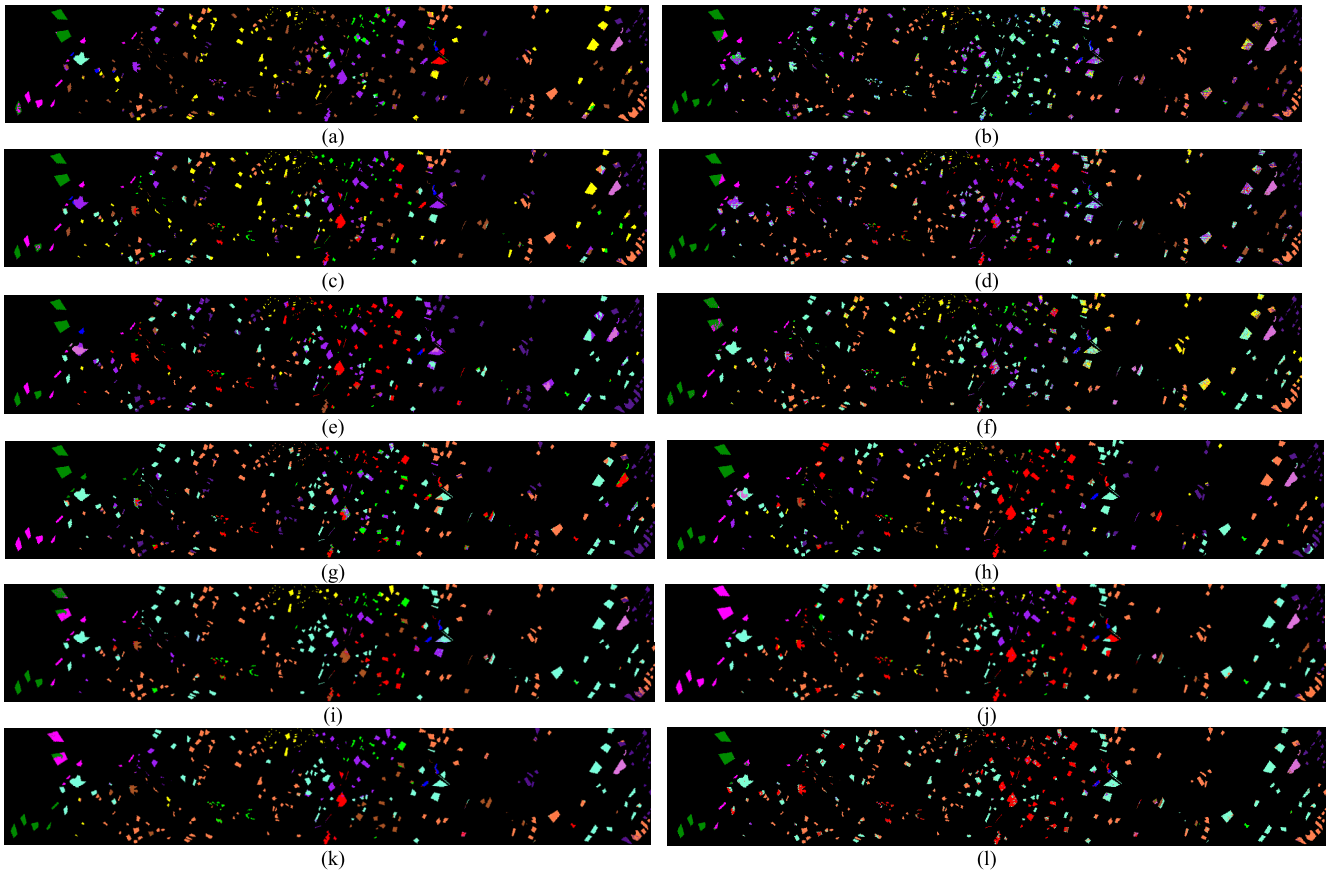


Fig. 8. Classification maps of different methods on Dioni dataset. (a) ASSMN. (b) GCN. (c) DAS-GCN. (d) SGC. (e) MDGCN. (f) GCGCN. (g) SSAPGCN. (h) CEGCN. (i) WFCG. (j) HybridFormer. (k) MARP. (l) CIGCS.

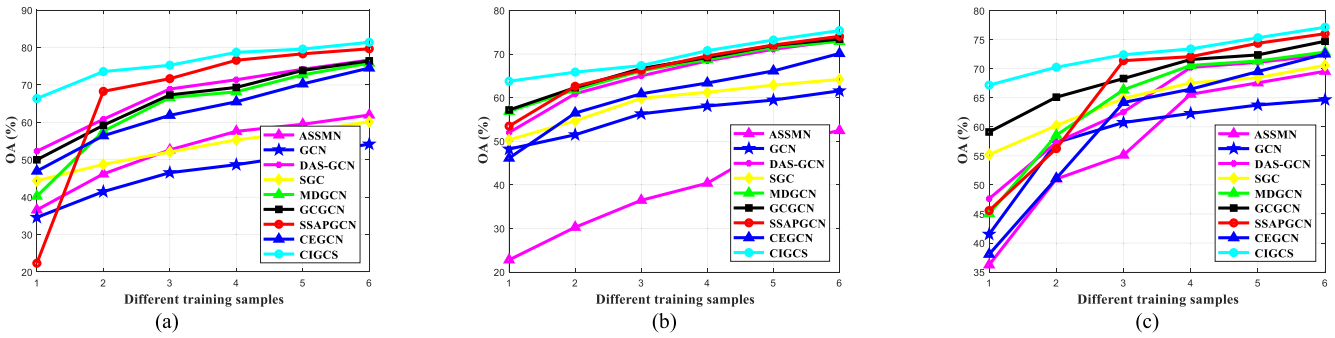


Fig. 9. Overall classification performance of different classification methods with different training samples. (a) Indian Pines. (b) Houston. (c) Dioni.

structure. From Fig. 10, the parameter φ is usually to be large on three datasets, which denotes that the sum of the C smallest eigenvalues of the similarity matrix $\boldsymbol{S}$ is close to zero and demonstrates the reasonability of Laplacian regularizer in CIGCS method. The parameter φ was set to 1 for three datasets. Besides, the parameter λ was set 0.1 for Indian Pines and Dioni datasets, and the parameter λ was set 10. The main reason for this phenomenon is that the restriction of parameter φ will be larger because the distribution of the class regions in Houston is usually small and the local Euclidean distance of the spatial–spectral information will be small. In fact, the parameter ε controls the degree of feature smoothing for imbalanced class regions. The variation of parameter ε ranges from [0.000001, 1]. Generally, the variation of parameter ε causes the change in the precision, which shows a trend of first rising and then falling. In order to obtain effective feature smoothing, the value of parameter ε is generally to be small due to the consistency of intraclass feature smoothing. From Fig. 10(d), the classification result of the Indian Pines data is best when setting the value of parameter ε as 0.0001. Similarly, the value of parameter ε on Houston 2013 and Dioni datasets can be set as 0.001 and 0.001, respectively, in order to obtain the best classification results.

G. Analysis of Clustering Performance

Considering that the quality of the feature smoothing is beneficial to obtain the convinced classification results, the

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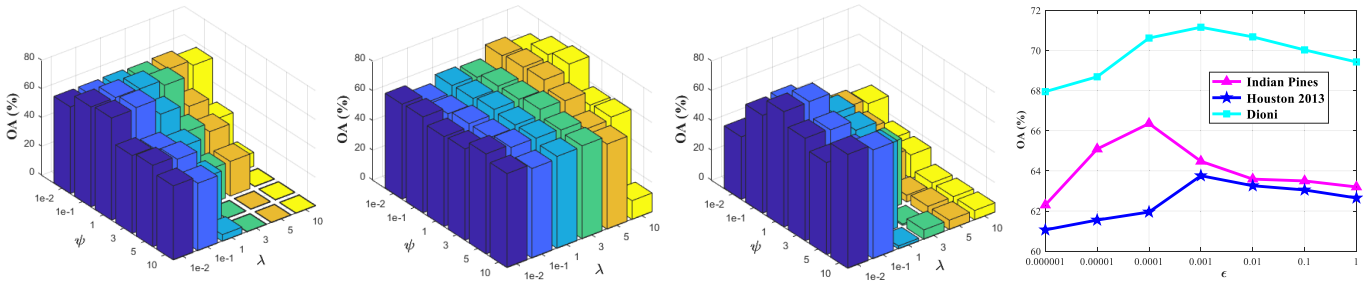


Fig. 10. Classification performance of different parameters in the proposed CIGCS method. (a) Indian Pines. (b) Houston. (c) Dioni. (d) Parameter ε.

TABLE V
OVERALL CLASSIFICATION PERFORMANCE OF THE CIGCS WITH ABLATION STUDY ON THREE DATASETS

Indian Pines				
Methods	SGC	SGCB	SGCIM	CIGCS
OA	51.12(4.97)	61.33(6.42)	60.09(6.33)	**66.36**(5.85)
AA	60.49(4.60)	72.21(3.90)	75.03(3.60)	**80.88**(2.39)
KC	44.56(5.34)	56.62(6.64)	55.71(6.31)	**62.75**(5.83)
Houston 2013				
Methods	SGC	SGCB	SGCIM	CIGCS
OA	49.28(3.36)	58.20(4.56)	55.60(4.28)	**63.76**(3.22)
AA	55.10(2.94)	61.69(3.69)	58.43(3.41)	**67.10**(2.59)
KC	45.01(3.58)	54.90(4.98)	52.05(4.94)	**60.94**(3.46)
Dioni				
Methods	SGC	SGCB	SGCIM	CIGCS
OA	55.23(8.77)	64.04(6.61)	61.85(4.12)	**71.16**(2.87)
AA	63.86(6.38)	70.79(5.11)	69.61(3.37)	**78.84**(2.07)
KC	47.59(9.62)	57.66(6.41)	53.68(4.78)	**64.78**(3.50)

TABLE VI
CLUSTERING PERFORMANCE OF DIFFERENT METHODS ON THREE DATASETS

Indian Pines						
Methods	Original	SGC	GCGCN	HybridFormer	MARP	CIGCS
AC	36.24	48.02	50.37	56.81	57.24	**58.93**
NMI	42.35	46.95	49.65	57.14	58.12	**59.81**
Houston 2013						
Methods	Original	SGC	GCGCN	HybridFormer	MARP	CIGCS
AC	43.76	51.73	54.57	58.21	58.56	**59.76**
NMI	43.90	50.01	54.85	58.43	58.93	**60.58**
Dioni						
Methods	Original	SGC	GCGCN	HybridFormer	MARP	CIGCS
AC	35.94	55.76	55.96	56.31	56.56	**57.84**
NMI	48.75	52.80	54.61	56.45	57.12	**59.50**

*AC denotes the clustering accuracy, and NMI denotes normalized mutual information. The closer they are to 100 %, the better their clustering performance is.

related experiments with the added analysis of the clustering tasks with different feature smoothing methods are conducted to be observed in Table VI. From Table VI, it can be observed that the clustering result via the feature smoothing of the CIGCS method is superior to the other feature smoothing methods with ac and NMI indicators. Meanwhile, the ac and NMI as evaluation indicators are used to evaluate the clustering performance. The closer they are to 100%, the better their clustering performance. The "origin" denotes the direct clustering result of original HSI datasets without any preprocessing methods. Compared to the clustering results of original HSI, the clustering results via the SGC, GCGCN, HybridFormer, MAPR, and CIGCS methods demonstrate better clustering performance because the smoothed features via graph convolution are beneficial to the downstream clustering task. The clustering results of smoothed features via GCGCN, HybridFormer, and MARP methods are superior to that of the SGC method because the global smoothed features are beneficial to the clustering task compared to the local smoothed feature via SGC method. Next, the clustering result of smoothed feature via the CIGCS method is better than that of the other method because the CIGCS method is able to achieve adaptive feature smoothing for imbalanced class regions instead of the local and global feature smoothing methods.

H. Analysis of Adaptive Hops Neighbors

In order to observe whether the proposed CIGCS method is able to achieve the adaptive neighboring aggregation for each semantic class, the related experiments with the analysis of the adaptive neighboring aggregation are conducted to be shown in Table VII. From Table VII, the number of pixels within semantic class is given, and the k_i-hops neighbors information for achieving the effective feature smoothing are also obtained by the CIGCS method for each class c_i at the class levels. The proposed CIGCS method achieves adaptive neighboring information aggregation for effective feature smoothing instead of adopting fixed propagation paradigm for all classes in HSIs. For example, a few classes, such as "Alfalfa," "Grass-pasture-mowed," "Oats," and "Stone Steel-Towers" with limited contextual information (i.e., sparse connectivity graph structure), usually capture high-order hops (i.e., 18, 6, 8, 15 hops, respectively) neighbors to achieve neighboring information aggregation for effective feature smoothing in Indian Pines data. Most GCN-based methods only capture fixed two or three hops neighbors to achieve information aggregation. In the Houston 2013 dataset, a few classes, such as "Parking lot 1" and "Parking lot 2" equipped with similar spectral information, often obtain local graph structure in order to guarantee its discriminability. Thus, they usually capture high-order hops neighbors to achieve effective feature smoothing of intraclass pixels. Similarly, classes with "Tennis court" and "Running track" usually capture high-order hops neighbors (that are 13 and 14 hops, respectively) because the local graph structure is beneficial to discriminate its similar spectral information. For

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TABLE VII
ADAPTIVE HOPS FOR INFORMATION AGGREGATION OF EACH SEMANTIC CLASS ON INDIAN PINES, HOUSTON, AND DIONI DATASETS

Indian Pines				Houston 2013				Dioni			
Class	Name	Pixel number	Hops	Class	Name	Pixel number	Hops	Class	Name	Pixel number	Hops
1	Alfalfa	46	18	1	Healthy grass	1251	2	1	Dense Urban Fabric	1262	2
2	Corn-notill	1428	3	2	Stressed grass	1254	1	2	Mineral Extraction Sites	204	1
3	Corn-mintill	830	2	3	Synthetic grass	697	2	3	Non Irrigated Arable Land	614	2
4	Corn	237	3	4	Tree	1244	3	4	Fruit Trees	150	3
5	Grass-pasture	483	2	5	Soil	1242	4	5	Olive Groves	1768	4
6	Grass-trees	730	5	6	Water	325	5	6	Coniferous Forest	361	5
7	Grass-pasture-mowed	28	6	7	Residential	1268	6	7	Dense Sclerophyllous Vegetation	5035	6
8	Hay-windrowed	478	7	8	Commercial	1244	7	8	Sparce Sclerophyllour Vegetation	6374	7
9	Oats	20	8	9	Road	1252	8	9	Sparcely Vegetated Areas	1754	8
10	Soybean-notill	972	9	10	Highway	1227	9	10	Rock and Sand	492	9
11	Soybean-mintill	2455	10	11	Railway	1235	10	11	Water	1612	10
12	Soybean-clean	593	11	12	Parking lot 1	1233	11	12	Coastal Water	398	11
13	Wheat	205	12	13	Parking lot 2	469	12				
14	Woods	1265	13	14	Tennis court	428	13				
15	Buildings-Grass-Trees-Drives	386	14	15	Running track	660	14				
16	Stone-Steel-Towers	93	15								

the Dioni dataset, a few classes with "Dense Urban Fabric," "Mineral Extraction Sites," "Non Irrigated Arable Land," and "Fruit Trees" only capture low-order hops neighbors to achieve effective feature smoothing of intraclass pixels because their spectral information is different to a certain degree, and they are able to be described via global graph structure. Thus, their effective feature smoothing can be achieved by the information aggregation of low-order neighbors. Moreover, there is an interesting phenomenon that the proposed CIGCS method is able to aggregate 18 hops neighbors alleviating the oversmoothing issue of most GCN-based methods to a certain degree due to its decoupled propagation paradigm of adaptive feature smoothing with imbalanced semantic classes.

VI. CONCLUSION

In this article, the effective CIGCS method was proposed to deal with class-imbalanced problem in graph convolution smoothing, which improves the quality of the feature smoothing in GCN-based methods for HSI classification. The CIGCS method mainly includes two techniques: the semantic block-diagonal graph structure is to describe imbalanced class distribution and the CIGCS operator is to achieve adaptive feature smoothing for each class region. Besides, the adaptive hops for neighboring information aggregation can be guaranteed by the difference of Euclidean distance of smoothed features within each block-diagonal structure and its theoretical upper bound. Experimental results on three real hyperspectral datasets demonstrate the effectiveness of the CIGCS method based on OA, KC, and AA indexes. Compared with the other popular GCN-based methods, the CIGCS method achieves the best classification results by considering the class-imbalanced problem in graph convolution smoothing. In the future, the CIGCS method can be further improved. For example, how to choose the oversmoothing measurement for each semantic class achieving fine feature smoothing will be a focus of future research.

APPENDIX

Proof for Theorem 1

Recall that the optimal K_r-hops neighbors for imbalanced class regions described with block-diagonal structure b_r can be defined as

$$K_r(b_r) = \min\left\{ \| \operatorname{intra}\left(b_r, K_r^{\infty}\right) - \operatorname{intra}(b_r, K_r) \| < \varepsilon \right\}.$$

In terms of (15), the essence of the smoothed samples' distance by K_r and K_r^{∞}-hops neighbors in intra(b_r, K_r) and intra(b_r, K_r^{∞}) is from the original feature matrix $\boldsymbol{X}$, i.e., $\hat{\boldsymbol{X}}_{b_r} = \boldsymbol{S}_{b_r}^{K_r} \cdot \boldsymbol{X}_{b_r}$. Consequently, the above formula for each block-diagonal structure b_r can be rewritten as shown in the equation at the top of the next page.

Considering that the adaptive the aggregation of neighboring information is close to the graph structure, that is, the choice of K_r and K_r^{∞}-hops neighbors for imbalanced class regions, the above formula can be simplified as the following problem:

$$K_r(b_r, \varepsilon) = \min\left\{ \left\| \boldsymbol{S}_{b_r}^{K_r^{\infty}} - \boldsymbol{S}_{b_r}^{K_r} \right\|_2^2 < \varepsilon \right\}.$$

Inspired by [51] and [52], the obtained block-diagonal structure can guarantee to be connected in intraclass pixels, which is necessary condition. Given the similarity matrix $\boldsymbol{S}_{b_r}$ for each block-diagonal structure, we denote $\boldsymbol{D}_{b_r}^{-1}\boldsymbol{S}_{b_r}$ as a transition matrix $\mathbf{P}$ where $\boldsymbol{D}_{b_r}^{-1}$ denotes the degree matrix of the similarity matrix $\boldsymbol{S}_{b_r}$. According to [51], for any initial distribution π_0

$$\tilde{\pi}(\pi_0)_i = \frac{1}{N_{b_r}} \sum_{j \in b_r} \mathbf{P}_{ji} * \sum_{j \in b_r} \pi_{0j}$$

where $\tilde{\pi}_i$ denotes the ith component of transformed feature distribution $\tilde{\pi}(\pi_0)$, i.e., $\tilde{\pi}(\pi_0) = \lim_{K_r \to \infty} \pi_0 \mathbf{P}^{K_r}$, and N_{b_r} denotes the number of nodes in block-diagonal structure b_r. From the above formula, the transformed feature distribution $\tilde{\pi}(\pi_0)$ can be replaced by $\tilde{\pi}$ due to $\tilde{\pi}$ is independent to π_0 to a certain degree. Based on the analysis, the adaptive mixing time with initial distribution is defined as

$$T(\pi_0, \varepsilon) = \min\left\{ t : \| \tilde{\pi} - \pi_0 \mathbf{P}^t \|_2 < \varepsilon \right\}.$$

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$$\min\left\{\begin{array}{l}\| \frac{1}{|b_r|(|b_r|)}\sum_{\boldsymbol{x}_i,\boldsymbol{x}_j\in b_r}\| \left(\boldsymbol{S}_{b_r}^{K_r^\infty}\cdot\boldsymbol{X}_{b_r}\right)_i^{K_r^\infty}-\left(\boldsymbol{S}_{b_r}^{K_r^\infty}\cdot\boldsymbol{X}_{b_r}\right)_j^{K_r^\infty}\|_2^2- \\ \frac{1}{|b_r|(|b_r|)}\sum_{\boldsymbol{x}_i,\boldsymbol{x}_j\in b_r}\| \left(\boldsymbol{S}_{b_r}^{K_r}\cdot\boldsymbol{X}_{b_r}\right)_i^{K_r}-\left(\boldsymbol{S}_{b_r}^{K_r}\cdot\boldsymbol{X}_{b_r}\right)_j^{K_r}\|_2^2\|<\varepsilon\end{array}\right\}$$
$$=\min\left\{\begin{array}{l}\| \frac{1}{|b_r|(|b_r|)}\|\boldsymbol{S}_{b_r}^{K_r^\infty}\|_2^2\sum_{\boldsymbol{x}_i,\boldsymbol{x}_j\in b_r}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2- \\ \frac{1}{|b_r|(|b_r|)}\|\boldsymbol{S}_{b_r}^{K_r}\|_2^2\sum_{\boldsymbol{x}_i,\ \boldsymbol{x}_j\in b_r}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2\|<\varepsilon\end{array}\right\}.$$

To better analyze the distribution of each block diagonal, we need to consider the impact of each node in each block separately, let $\pi_0 = e_i$ be a one-hot vector with the ith component equal to 1, and the other components equal to 0. According to [51], we can derive Lemma 1.

Lemma 1: Following the low-pass filtering property [53], [54] with self-loops in a constructed graph structure, given the degree matrix $\tilde{d}_j$ and $\tilde{d}_i$ of nodes, we can derive the following lemma:

$$\left|\left(e_i\mathbf{P}^t\right)_j-\tilde{\pi}_j\right|\le\sqrt{\frac{\tilde{d}_j}{\tilde{d}_i}}\lambda_2^t$$

where λ_2 is the second large eigenvalue of transition matrix $\mathbf{P}$, $\tilde{d}_i = d_i + 1$, $\tilde{d}_j = d_j + 1$, and $(e_i\mathbf{P}^t)_j$ symbols the jth element of $e_i\mathbf{P}^t$. We denote $e_i\mathbf{P}^t$ as $\pi_i(t)$, and then,

$$\begin{aligned}\|\tilde{\pi}-\pi_i(t)\|_2^2&=\sum_{j=1}^{N_{b_r}}\left(\tilde{\pi}_j-\pi_i(t)_j\right)^2\\&\le\frac{\sum_{j=1}^{N_{b_r}}\tilde{d}_j}{\tilde{d}_i}\lambda_2^{2t}=\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}\lambda_2^{2t}\end{aligned}$$

where M_{b_r} and N_{b_r} denote the number of edges and nodes, respectively. Next, we can obtain the following result:

$$\|\tilde{\pi}-\pi_i(t)\|_2^2\le\sqrt{\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}}\lambda_2^t.$$

Now, let

$$\varepsilon\le\sqrt{\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}}\lambda_2^t.$$

Consequently, there exists

$$T(e_i,\varepsilon)\le\sqrt{\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}}\lambda_2^t.$$

Inspired by $\tilde{\pi}(\pi_0) = \lim_{K_r\to\infty}\pi_0\mathbf{P}^{K_r}$, we can derive the following result:

$$\|e_i\mathbf{P}^\infty-e_i\mathbf{P}^{K_r}\|_2^2\le\sqrt{\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}}\lambda_2^{K_r}.$$

Since $\boldsymbol{S}_{b_r}^{K_r}=e_i\mathbf{P}^{K_r}$ and $\boldsymbol{S}_{b_r}^{K_r^\infty}=e_i\mathbf{P}^\infty$, we have

$$\|\boldsymbol{S}_{b_r}^{K_r^\infty}-\boldsymbol{S}_{b_r}^{K_r}\|_2^2\le\sqrt{\frac{2M_{b_r}+N_{b_r}}{\tilde{d}_i}}\lambda_2^{K_r}.$$

Recall the choice of adaptive neighbors for imbalanced class regions, i.e., $K_r(b_r,\varepsilon)=\min\left\{\left\|\boldsymbol{S}_{b_r}^{K_r^\infty}-\boldsymbol{S}_{b_r}^{K_r}\right\|_2^2<\varepsilon\right\}$, we can let $\varepsilon=((2M_{b_r}+N_{b_r}/\tilde{d}_i))^{1/2}\lambda_2^{K_r}$. Then, we take the logarithm for $\varepsilon=((2M_{b_r}+N_{b_r}/\tilde{d}_i))^{1/2}\lambda_2^{K_r}$, and there exists

$$K_r(b_r,\varepsilon)\le\log_{\lambda_2}\left(\varepsilon\sqrt{\frac{\tilde{d}_i}{2M_{b_r}+N_{b_r}}}\right).$$

The proof is completed. From the above analysis, we can conclude that the node that has a lager degree may need to aggregate few hops neighbors to achieve information propagation and the node that has a smaller degree may need to aggregate more hops neighbors to achieve information propagation.

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